

Identifying invertebrate indicators for streamflow duration assessments in forested headwater streams

Ken M. Fritz^{1,7}, Roxolana O. Kashuba^{2,8}, Gregory J. Pond^{3,9}, Jay R. Christensen^{1,10}, Laurie C. Alexander^{2,11}, Benjamin J. Washington^{2,4,12}, Brent R. Johnson^{1,13}, David M. Walters^{5,14}, William T. Thoeny^{6,15}, and Paul C. Weaver^{1,16}

¹Office of Research and Development, United States Environmental Protection Agency, 26 West Martin Luther King Drive, Cincinnati, Ohio 45268 USA

²Office of Research and Development, United States Environmental Protection Agency, 1200 Pennsylvania Avenue Northwest, Washington, DC 20460 USA

³Region 3, United States Environmental Protection Agency, 1060 Chapline Street Suite 303, Wheeling, West Virginia 26003 USA

⁴Verisk Analytics, 545 Washington Boulevard, Jersey City, New Jersey 07310 USA

⁵US Geological Survey, Columbia Environmental Research Center, 4200 East New Haven Road, Columbia, Missouri 65201 USA

⁶Pegasus Technical Services, c/o United States Environmental Protection Agency, 26 West Martin Luther King Drive, Cincinnati, Ohio 45268 USA

Abstract: Streamflow-duration assessment methods (SDAMs) are rapid, indicator-based tools for classifying streamflow duration (e.g., intermittent vs perennial flow) at the reach scale. Indicators are easily assessed stream properties used as surrogates of flow duration, which is too resource intensive to measure directly for many reaches. Invertebrates are commonly used as SDAM indicators because many are not highly mobile, and different species have life stages that require flow for different durations and times of the year. The objectives of this study were to 1) identify invertebrate taxa that can be used as SDAM indicators to distinguish between stream reaches having intermittent and perennial flow, 2) to compare indicator strength across different taxonomic and numeric resolutions, and 3) to assess the relative importance of season and habitat type on the ability of invertebrates to predict streamflow-duration class. We used 2 methods, random forest models and indicator species analysis, to analyze aquatic and terrestrial invertebrate data (presence/absence, density, and biomass) at the family and genus levels from 370 samples collected from both erosional and depositional habitats during both wet and dry seasons. In total, 36 intermittent and 53 perennial reaches were sampled along 31 forested headwater streams in 4 level II ecoregions across the United States. Random forest models for family- and genus-level datasets had stream classification accuracy ranging from 88.9 to 93.2%, with slightly higher accuracy for density than for presence/absence and biomass datasets. Season (wet/dry) tended to be a stronger predictor of streamflow-duration class than habitat (erosional/depositional). Many taxa at the family (58.8%) and genus level (61.6%) were collected from both intermittent and perennial reaches, and most taxa that were exclusive to 1 streamflow-duration class were rarely collected. However, 23 family-level or higher taxa (20 aquatic and 3 terrestrial) and 44 aquatic genera were identified as potential indicators of streamflow-duration class for forested headwater streams. The utility of the potential indicators varied across level II ecoregions in part because of representation of intermittent and perennial reaches in the dataset but also because of variable ecological responses to drying among species. Aquatic invertebrates have been an important field indicator of perennial reaches in existing SDAMs, but our findings highlight how including aquatic and terrestrial invertebrates as indicators of intermittent reaches can further maximize the data collected for streamflow-duration classifications.

Key words: intermittent, perennial, invertebrate, indicators, flow classification, taxonomic resolution, numerical resolution

Natural processes and anthropogenic controls influence flow duration in streams (i.e., when and for how long channels carry moving surface water) across spatiotemporal scales

(Costigan et al. 2016, Shanafield et al. 2021). Streamflow-duration classification has been used to quantify effects of climate variability on stream mapping accuracy (Hafen

E-mail addresses: ⁷fritz.ken@epa.gov; ⁸kashuba.roxolana@epa.gov; ⁹pond.greg@epa.gov; ¹⁰christensen.jay@epa.gov; ¹¹alexander.laurie@epa.gov; ¹²bwash31892@gmail.com; ¹³johnson.brent@epa.gov; ¹⁴waltersd@usgs.gov; ¹⁵Retired, btthoeny@gmail.com; ¹⁶weaver.paul@epa.gov

Received 15 August 2022; Accepted 1 January 2023; Published online 5 July 2023. Associate Editor, Daren Carlisle.

Freshwater Science, volume 42, number 3, September 2023. © 2023 The Society for Freshwater Science. All rights reserved. Published by The University of Chicago Press for the Society for Freshwater Science. <https://doi.org/10.1086/726081>

et al. 2020), to assess habitat condition or restoration success (Woelfle-Erskine 2017), and for water resource governance, including informing federal jurisdictional determinations under the United States Clean Water Act (Fritz et al. 2017). Many different terms have been used to describe stream reaches by their flow duration (Busch et al. 2020), but perennial and intermittent are among the most common. Perennial reaches have temporally continuous surface flow and do not experience recurrent drying. Intermittent reaches experience surface flow for only part of the year. Headwater stream reaches are commonly intermittent because seasonal changes of groundwater elevation can cause the expression of surface flow to propagate upstream from perennial downstream reaches, expand downstream from perennial springs, or coalesce from both directions (Shanafield et al. 2021). The intent of streamflow-duration classification is to represent the typical regime over many years, even though the specific number of days a stream reach dries or carries surface streamflow may vary from year to year. As a consequence, the streamflow-duration class of a reach does not change from year to year.

Long-term data from ground and remote-sensing surveys (e.g., Hooshyar et al. 2015, Allen et al. 2019) and continuous monitoring with stream gauges or other devices (e.g., Poff 1996, Chapin et al. 2014, Hammond et al. 2021) can be used to directly classify reaches or in conjunction with hydrological modeling to predict streamflow duration across networks within catchments (Ward et al. 2018, Mahoney et al. 2023). However, most intermittent headwater streams are poorly represented on maps (van Meerveld et al. 2020), and direct streamflow monitoring and associated modeling has focused disproportionately on large perennial rivers draining the most densely populated watersheds (Messager et al. 2021, Krabbenhoft et al. 2022). The spatial extent of headwater streams is vast, and image resolution, cloud cover, and density of tree canopies can each limit application of remote sensing to characterize flow in headwater streams of many regions. Thus, monitoring, modeling, and remote-sensing approaches for streamflow classification are typically unavailable when data to support management decisions (e.g., water-quality standards, riparian buffer widths, regulatory jurisdiction) are needed for specific headwater reaches within a network. Streamflow-duration assessment methods (SDAMs) are rapid assessments that use indicators to classify streamflow duration at the reach scale (Fritz et al. 2020). SDAM indices or models use physical and biological indicators as surrogates for direct measurement of stream flow over time. Indicators can control or respond to—and therefore reflect—streamflow duration.

Aquatic invertebrates are among the most well-studied biological indicators in streams and have been widely used to assess water quality and, more generally, habitat condition (Rosenberg and Resh 1993). Their natural history and ecology have been among the most studied topics in intermittent streams (Williams 2006, Leigh et al. 2016). Drying is

an especially strong factor controlling the distribution, abundance, and diversity of aquatic invertebrates in streams (Stubbington et al. 2017). Many taxa have adaptations to resist, avoid, and recover quickly following periodic drying, but others without these adaptations are largely restricted to reaches with perennial flow (Davis et al. 2017, Cartwright et al. 2020). Aquatic invertebrates are commonly used as indicators in SDAMs (NCDWQ 2010, Nadeau et al. 2015, Mazor et al. 2021a, b). Although their presence in aquatic samples is typically assumed to be accidental, terrestrial and semiterrestrial invertebrates also inhabit a variety of intermittent stream habitats, even in the presence of streamflow (Corti and Datry 2012). As with aquatic invertebrates, the abundance and composition of terrestrial assemblages in streams depend on flow regime and vary in response to seasonal changes in habitat quality, quantity, and connectivity (Steward et al. 2022). Most research to date has focused on the contributions of terrestrial and semiterrestrial invertebrates to stream food webs and organic matter transport (Courtwright and May 2013, Rosado et al. 2015), but their contributions to the composition and diversity of intermittent stream communities and their utility as possible indicators of streamflow duration have been less studied (Moon 1956, Sánchez-Montoya et al. 2016).

We had 3 primary objectives. The 1st was to identify those invertebrate taxa (aquatic and terrestrial) that best discriminate between intermittent and perennial reaches and thus could serve as SDAM indicators for small, forested streams. Our 2nd objective was to determine if the numeric resolution (i.e., presence/absence, density, biomass) and taxonomic resolution (genus vs family) used to characterize invertebrate taxa affects classification accuracy. Answering this question would inform us whether field- or laboratory-based enumerations and identifications are needed for SDAM application. Our 3rd objective was to assess the relative influence of season (wet vs dry) and habitat (depositional vs erosional) on classification accuracy to optimize SDAM data collection.

METHODS

To address our research objectives, we conducted a field study that compared the invertebrate assemblage composition at 39 intermittent reaches and 51 perennial reaches in 31 forested headwater streams distributed across 10 mesic forests. We sampled invertebrate assemblages over consecutive wet and dry seasons for 1 or 2 y. At each visit from each reach, we collected 2 to 6 individual core samples from erosional and depositional habitats with visible surface water. Individual invertebrate samples were composited by habitat for each reach and visit, so there were a total of 132 composited samples from intermittent reaches and 238 composited samples from perennial reaches. We analyzed our data using a machine-learning approach and indicator species analysis (ISA) to identify potential invertebrate indicators of

streamflow-duration class. We also used machine learning to 1) identify models, built with different numeric and taxonomic resolutions of the invertebrate samples, with the highest streamflow-duration classification accuracy and 2) compare the relative importance of sample habitat (erosional and depositional) and season (wet and dry) on streamflow-duration classification.

Study design

We selected 10 mesic forests for study, spanning 4 level II ecoregions (Omernik 1995) and 10 different states across the United States (Table 1, Fig. 1). We sampled 31 headwater streams (1–4/forest) over consecutive wet (April and May) and dry (August and September) seasons for 1 or 2 y (2–4×/reach). Four forests (17 streams in ecoregions 8.3 and 8.4) were sampled over 2 consecutive y (2003, 2004), whereas 6 forests (14 streams in ecoregions 5.3, 6.2, 8.3, and 8.4) were sampled in April and May 2005 and August and September 2004 or 2005 (1 y/forest). The 4 forests sampled over 2 consecutive y were near to one another so were more logistically convenient to sample twice compared with the other 6 forests. Streams drained catchments with >90% forest cover and had drainage areas ranging from 0.07 to 293.4 ha (see Fritz et al. 2008 for more details about the study streams). No existing hydrologic data existed for any of the studied headwater stream reaches, so we identified reaches for sampling based on longitudinal position (or drainage area) along streams to capture a wide range in streamflow duration. Except in losing systems, streamflow duration is expected to increase with drainage area because it increases storage capacity and recharge potential. Along each stream, we identified 3 to 4 reaches (30 m each) for sampling. The number of reaches varied because stream length differed among study streams. We scored flow status (continuous

surface flow, interstitial flow, isolated pools, or dry) on each visit (for each reach on consecutive wet and dry seasons for 1 or 2 y). To improve accuracy of classification of reaches as intermittent or perennial, we classified reaches in 2 different ways. First, we classified intermittent reaches as those with isolated pools or that were dry during ≥ 1 visits but were flowing (continuous surface flow or interstitial) on ≥ 1 visit. We classified perennial reaches as those that were flowing on every visit. This classification identified 39 intermittent and 51 perennial reaches. Second, we used a subset of reaches (54/90) that had continuous data from electrical resistance data loggers to check the accuracy of our field classifications. The data loggers recorded when reaches dried and when they resumed flow, so the timing, duration, and frequency of flow could be calculated (Fritz et al. 2006). Data loggers were deployed in the channel thalweg in erosional habitat. Using these data, we classified reaches that were flowing 10 to 90% of the year as intermittent and reaches that were flowing >90% of the year as perennial (Hedman and Osterkamp 1982). Classification based on site visits and data loggers agreed in 85.2% of cases (88% agreement for intermittent reaches and 82.8% for perennial reaches).

Invertebrate collection

We sampled for invertebrates in habitats with visible surface water along each reach and characterized the habitats sampled as either erosional (e.g., riffle) or depositional (e.g., pool) during each visit. We used a 19-L bucket with the bottom removed and a 250- μ m-mesh hand net to collect 2 to 6 individual core samples to a depth of ≤ 10 cm (core volume: 530 cm³) across multiple units of each habitat type along the thalweg (deepest flow path) of each reach. We collected invertebrates by hand from coarse substrate within the bucket, stirred the remaining sediment, and used the hand

Table 1. Study locations in 10 states across the United States. Shown are level II ecoregion names and codes, study years, no. of streams/location, no. of intermittent reaches and corresponding no. of composite samples, and no. of perennial reaches and corresponding no. of composite samples.

Study location	Level II ecoregion			Streams	Intermittent reaches		Perennial reaches	
	Name	Code	Year		Reaches	Samples	Reaches	Samples
Illinois	Southeastern USA Plains	8.3	2004–2005	4	10	19	1	4
Indiana	Southeastern USA Plains	8.3	2003–2004	4	13	58	0	0
Kentucky	Ozark, Ouachita-Appalachian Forests	8.4	2003–2004	5	8	37	6	46
South-central Ohio	Ozark, Ouachita-Appalachian Forests	8.4	2003–2004	4	0	0	9	52
Southeastern Ohio	Ozark, Ouachita-Appalachian Forests	8.4	2003–2004	4	5	11	8	45
New Hampshire	Atlantic Highlands	5.3	2005	1	1	3	2	5
New York	Atlantic Highlands	5.3	2004–2005	2	0	0	6	15
Vermont	Atlantic Highlands	5.3	2005	1	1	2	2	8
Washington	Western Cordillera	6.2	2004–2005	2	1	2	5	19
West Virginia	Ozark, Ouachita-Appalachian Forests	8.4	2004–2005	4	0	0	12	44
Total				31	39	132	51	238

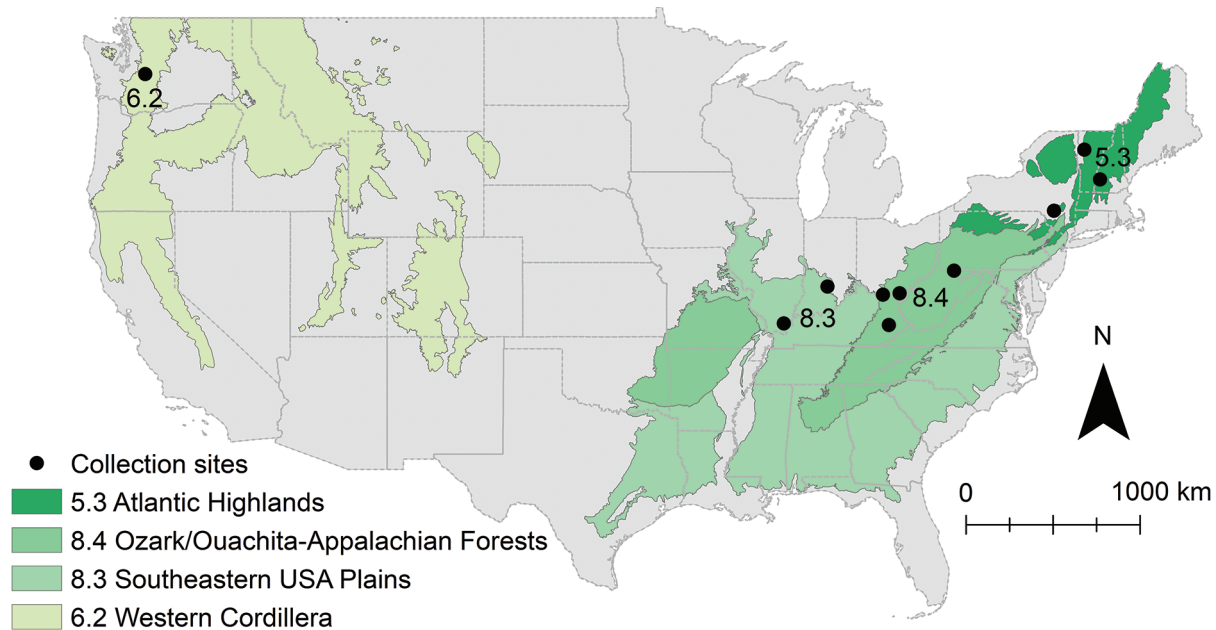


Figure 1. Map of 10 study forests with invertebrate collection sites in 4 level II ecoregions across the United States. Numbers correspond with level II ecoregions as designated by the Commission for Environmental Cooperation.

net to collect invertebrates suspended in the water (see Fritz et al. 2006 for a detailed sampling description). We retained all terrestrial invertebrates collected in the core samples. We preserved samples in the field with 95% ethanol and transported them to the laboratory. If a reach had no visible surface water during a visit, it was not sampled at that time.

We processed and identified the invertebrate samples in the lab following Klemm et al. (1990). The invertebrates from most samples (1466/1559) were completely sorted, and the remaining 93 were subsampled. For the subsamples, we used a 2-phased processing method. First, we searched the entire sample for large and rare organisms, and then we used a 500-ind. fixed count to subsample the remaining fraction (Caton 1991, Moulton et al. 2000). We enumerated the invertebrates and identified them to the lowest possible taxon (typically genus for aquatic taxa and family or order for terrestrial taxa) using taxonomic keys (e.g., Stehr 1987, 1991, Merritt and Cummins 1996, Epler 2001, Thorp and Covich 2001) and expert knowledge. We then assigned the invertebrates to aquatic or terrestrial categories by life stage and taxonomic identity and measured their lengths (tip of head to end of abdomen) to the nearest 1 mm to quantify body size. We used published length–mass equations (e.g., Edwards 1967, Nolte 1990, Sample et al. 1993, Benke et al. 1999) to convert body size measurements to biomass (mg ash-free dry mass/m²). We used equations associated with the identified taxonomic level when available and equations applicable to the lowest possible taxonomic level when published equations at the identified taxonomic level were unavailable. When >1 published equation was available for a taxon, we used the equation derived from in-

dividuals collected closest in geographic proximity to each sampled site. For each taxon, we composited the density (ind./m²) and biomass (mg ash-free dry mass/m²) estimates across those samples collected from each habitat type in a reach during a visit. This process resulted in a total of 370 composited samples used in subsequent analyses (Table 2).

Identifying invertebrate indicators

We calculated the total density, total biomass, aquatic family richness, terrestrial family richness, and aquatic genus richness for each composite sample. Richness for aquatic family, terrestrial family, and aquatic genus was calculated as the number of aquatic families, terrestrial families, and aquatic genera, respectively, within a composite sample. We used Wilcoxon rank-sum tests in R (version 4.1.3; R Project for Statistical Computing, Vienna, Austria) to compare total density, total biomass, aquatic family richness, terrestrial family richness, and aquatic genus richness between intermittent and perennial reaches. We used 2 approaches, random forest (RF) classification (Cutler et al. 2007) and ISA

Table 2. Number of composited invertebrate samples collected from intermittent and perennial reaches during wet and dry seasons from erosional and depositional habitats.

Season	Habitat type	Intermittent	Perennial
Wet	Erosional	52	57
Wet	Depositional	57	54
Dry	Erosional	4	61
Dry	Depositional	19	66

(De Cáceres and Legendre 2009), to identify potential genus- and family-level invertebrate indicators (i.e., those taxa over-represented in either intermittent or perennial reaches). We also used RF to 1) assess if either numeric (presence/absence, density, biomass) or taxonomic (family terrestrial and aquatic taxa, family aquatic only, genus aquatic only) resolutions influenced streamflow-duration classification accuracy and 2) assess the relative influence of sample habitat type (erosional and depositional) or season (wet and dry) on the accuracy of streamflow-duration classification. We compared the indicators identified between the RF approach and the ISA results to strengthen our confidence through consistency in the identified taxa being those best able to discriminate between the streamflow-duration classes at the family and genus taxonomic levels.

RF An RF is a type of enhanced bagging routine that fits many minimally correlated decision trees via recursive partitioning (Cutler et al. 2007). Correlation between decision trees in RF is reduced via bootstrapping and randomly subsampling the original data so that each split is determined using a random feature set. Finally, an RF uses ensemble learning across each of these decision trees, which we use in this study to identify indicators of streamflow-duration class (binary response variable: perennial or intermittent class). We conducted 9 separate RF analyses with presence/absence, density, and biomass predictors with combined aquatic and terrestrial family-level data, aquatic-only family-level data, and aquatic-only genus-level taxa to assess classification accuracy among different invertebrate occupancy estimators. In addition to data for each taxon, we included total aquatic and total terrestrial density or biomass as predictor variables in respective RF analyses that included aquatic taxa density, terrestrial taxa density, aquatic taxa biomass, and terrestrial taxa biomass. We also included habitat type (categorical variable: erosional or depositional) and season (categorical variable: dry or wet) in the models to assess their relative effect on classification accuracy. Our intent in including habitat type and season was to understand their relative importance for the predictive ability of invertebrate taxa (i.e., invertebrates may be more useful as indicators in one habitat or season than in another) rather than to interpret these variables directly as predictors of streamflow-duration class. To run the RF models, we used the *ranger* (version 0.14.1; Wright et al. 2022) and *randomForestExplainer* (version 0.10.1; Paluszynska et al. 2020) packages in R (version 4.0.2) to fit models with 500 decision trees. For each model, we used default out-of-bag settings (100% of data with replacement) to estimate classification accuracy (i.e., the % of out-of-bag samples incorrectly classified across all decision trees).

RF models are sensitive to imbalanced data because RF fitting methods bias toward correct classification of the majority class more so than of the minority class (Chen et al. 2004, More and Rana 2017). Here, we balanced the decision

tree by subsampling across streamflow-duration class during model fitting via 50/50%-weighted random sampling with replacement to prevent bias attributable to the dataset distribution of 64% ($n = 238$) of perennial stream samples and 36% ($n = 132$) of intermittent stream samples. Furthermore, samples were unevenly distributed across ecoregions: ~9% of samples from ecoregion 5.3, ~6% from ecoregion 6.2, ~22% from ecoregion 8.3, and ~64% from ecoregion 8.4 (Table 1). However, small sample sizes for some combinations of ecoregion and streamflow-duration class prevented concurrent balancing. Therefore, we present the indicator results by ecoregion to qualitatively evaluate differences among ecoregions.

To identify indicators that were predictive of streamflow-duration class for each of the 3 levels of numeric resolution, we ranked the taxa that most distinguished intermittent and perennial reaches in each model. We applied this procedure at each of the 3 levels of taxonomic resolution as well. Within a decision tree, recursive partitioning identifies splits in the data associated with different predictors. We used this information to rank predictors by their ability to separate perennial from intermittent reach samples. The 1st split in a decision tree is assigned a node depth of 0. A smaller node number (or lower node depth) typically denotes better class separation ability (Paluszynska et al. 2020). Across the 500 decision trees in each RF model, we calculated mean minimal node depth for each taxon predictor (based on only that predictor's earliest appearance in a decision tree if it appeared more than once). After verifying the linearity of relationships in family- and genus-level taxa ranks among RF models, we calculated Pearson's correlation coefficients for pairs of models to assess consistency among models for the 3 numeric resolutions. The units of mean minimal node depth are not comparable across RF models (i.e., they are only meaningful to compare predictors within a model). Therefore, to compare RF models built with data of different numeric resolution, we assigned a rank of 1 to taxa within each RF model with the lowest mean minimal node depth, a rank of 2 to the 2nd lowest, and so on. Then, for each taxon, we determined median rank across all 3 numeric resolutions. We plotted the median rank for each taxon across the 3 RF models of increasing numerical resolution in relation to the difference in % presence in samples from perennial and intermittent reaches calculated from the raw data. From these plots, we identified potential streamflow indicator taxa as those having low median rank node depths from RF models and >20% difference in presence between perennial and intermittent stream samples (i.e., taxa present in >20% more perennial stream samples than intermittent stream samples were identified as perennial indicators, and taxa present in >20% more intermittent stream samples than perennial stream samples were identified as intermittent indicators). We selected a threshold of $\pm 20\%$ during preliminary data analysis because this threshold identified taxa predictive of each streamflow-duration class, which

could be used in combination for SDAM metrics (e.g., total abundance of individuals of indicator taxa of perennial reaches, number of indicator taxa of intermittent reaches). Finally, we assessed the relative importance of sampling season and sample habitat on streamflow-duration classification by also plotting the median rank for season and habitat across the 3 RF models.

ISA ISA determines if ≥ 2 sets of samples differ in the relative abundances and occurrence frequencies of different taxa (De Cáceres and Legendre 2009). For this study, we used specificity and sensitivity to quantify affinity to streamflow-duration class. We conducted 2 separate ISAs with the aquatic and terrestrial family-level density data and genus-level aquatic taxa density data to identify indicator taxa for intermittent and perennial reaches. Specificity is the probability that a reach belongs to a streamflow-duration class given that a taxon occurred in a sample collected from the reach. Sensitivity is the mean relative number of individuals of a taxon in a sample collected from reaches in a streamflow-duration class. The square root of the product of specificity and sensitivity is the indicator value (IV) index. We then used the difference between the maximum observed IV for a streamflow-duration class and the mean IV generated from random permutations ($n = 999$) to derive a significance test for each taxon (De Cáceres and Legendre 2009). We identified taxa that had most ($IV > 0.5$) of their relative abundance and occurrences associated with 1 of the streamflow-duration classes as potential indicators. We ran ISA in R with the `multiplatt` function within the `indicspecies` package (version 1.7.8; De Cáceres et al. 2019).

RESULTS

We collected a total of 519,700 invertebrates, of which 439 genera, 121 families, and 25 higher-level taxa were aquatic and 133 families and 39 higher-level taxa were terrestrial. Of the 146 aquatic family- or higher-level taxa identified, 121 taxa occurred in >2 composite samples, whereas only 80/172 terrestrial family- or higher-level taxa occurred in >2 composite samples. Intermittent and perennial reaches did not differ in total invertebrate densities, total invertebrate biomass, aquatic family richness, or aquatic genus richness (Table 3). Terrestrial family richness was higher at intermittent reaches than at perennial reaches (Wilcoxon $W = 23,308$, $p < 0.001$).

Most taxa occurred in both intermittent and perennial reaches. Approximately 60% of taxa in both the combined aquatic plus terrestrial dataset of family- and higher-level taxa (58.8%) and the aquatic genus dataset (61.6%) occurred in both intermittent and perennial reaches. In addition, no taxa that occurred in $\geq 20\%$ of composite samples from a streamflow-duration class were exclusive to either intermittent or perennial reaches. Only 4 families and 18 genera that were exclusive to either perennial or intermittent

reaches occurred in >10 composite samples (Appendix S1 tabs 1–3). For example, although the mayfly genus *Siphonurus* (Siphonuridae) and stonefly genus *Perlesta* (Perlidae) were both exclusively collected from intermittent reaches, they were only observed in 9.8% (13/132 samples) and 11.4% (15/132 samples) of intermittent-reach samples. In contrast, there were 127 families (aquatic and terrestrial) and 200 aquatic genera that were exclusive to either perennial or intermittent reaches but occurred in <10 composite samples. Of these, 62% and 53% were singletons in the family-level and genus-level datasets, respectively.

RF classification accuracy

Overall classification accuracy of RF models (i.e., % of observed sample streamflow-duration classes correctly predicted by RF majority vote) were similar across numeric and taxonomic resolutions and ranged from 88.9 to 93.2% (Table 4). The maximum increase in classification accuracy by increasing numeric resolution from presence/absence to density or biomass in RF models was 2.5%. The maximum increase in classification accuracy by increasing taxonomic resolution from aquatic family-level taxonomy to aquatic genus-level taxonomy in RF models was 3.5%. All RF models had higher classification accuracy for perennial reaches than for intermittent reaches (Table 4).

The relative importance of taxonomic (family vs genus) and numeric (presence/absence, density, biomass) resolution in classification accuracy of RF models were similar among level II ecoregions (Table 4). The RF model using density and genus-level taxonomy had the highest classification accuracy for 3/4 level II ecoregions. The RF model using density and aquatic and terrestrial family taxonomy had the highest classification accuracy for ecoregion 8.4. Using genus-level taxonomy instead of family-level taxonomy resulted in greater improvements of classification accuracy than using density instead of presence/absence in ecoregions 6.2 and 8.3, but both changes made similar improvements in ecoregions 5.3 and 8.4. Using genus-level taxonomy instead of family-level taxonomy improved classification accuracy more than using biomass instead of presence/absence in ecoregions 5.3, 6.2, and 8.3, but improvements were comparable for ecoregion 8.4. Including terrestrial and aquatic families instead of only aquatic families did not improve classification accuracy as much as using density instead of presence/absence in ecoregions 5.3 and 6.2 and made comparable improvements in the other ecoregions.

RF indicator taxa

Family- and genus-level invertebrate indicators were identified from taxa included in RF models. Family taxa ranks were strongly correlated among RF models ($r \geq 0.89$), but correlations were weaker for genus taxa ranks ($r \geq 0.78$; Appendix S1 tab 4). Taxa with the best (lowest) median ranks

Table 3. Invertebrate metrics (mean $\pm$ SD, range) with results of Wilcoxon rank tests (W) between intermittent ($n = 132$) and perennial ($n = 238$) reaches. AFDM = ash-free dry mass.

Metric	Intermittent			Perennial			W	p
	Mean	SD	Range	Mean	SD	Range		
Total density (ind./m ²)	6902.6	4483.2	849.0–30,220.3	7377.4	5834.1	434.0–37,715.5	15,873	0.87
Total biomass (mg AFDM/m ²)	3514.9	2311.0	385.1–14,858.1	3459.2	2913.0	223.2–17,799.3	17,209	0.13
Aquatic family richness	24.4	5.7	9–39	25.6	6.5	10–47	14,106	0.10
Terrestrial family richness	6.1	3.9	0–20	3.2	2.2	0–10	23,308	<0.001
Aquatic genus richness	53.2	13.9	20–88	57.3	18.7	16–111	14,042	0.09

(i.e., Crangonyctidae, Elmidae, *Crangonyx*, *Lirceus*, *Stempellinella*, *Fridericia*) were predictive of streamflow-duration class regardless of numeric resolution used. Among the aquatic and terrestrial family-level invertebrate indicators identified through the RF models were 11 taxa associated with perennial flow and 10 taxa associated with intermittent flow (Table 5, Fig. 2A, Appendix S1 tab 2). All family-level indicators for perennial flow were for individuals with aquatic life stages, whereas there were 7 aquatic and 3 terrestrial indicators for intermittent flow. Perennial flow indicators included 10 insect and 1 crustacean family, whereas intermittent indicators included 6 insect, 2 crustacean, and 2 annelid taxonomic groups. The genus-level indicators included 18 taxa associated with perennial flow and 22 taxa associated with intermittent flow (Table 5, Fig. 2B, Appendix S1 tab 3). All the genera identified as perennial streamflow indicators were insects, including representatives from 6 orders and 11 families of aquatic insects. In contrast, intermittent flow indicators identified from the genus-level RF models included 3 crustacean and 3 annelid taxa as well as aquatic insects from 4 orders and 6 families. The perennial streamflow indicators included genera in 8 families that were also identified as perennial streamflow indicators, as

well as genera in 3 aquatic insect families (Chironomidae, Tipulidae, and Heptageniidae) that were not identified as family-level indicators. As identified in the family-level analyses, members of the order Harpacticoida and Megadrili (mixed taxonomic group that includes aquatic Lumbricidae and Sparganophilidae) were identified as intermittent flow indicators in the genus-level RF analyses (Table 5, Appendix S1 tabs 2, 3) along with members of 3 aquatic insect families (Dytiscidae, Lepidostomatidae, and Thremmatidae). Five families (Asellidae, Perlodidae, Chironomidae, Naididae, and Ceratopogonidae) had genera identified as genus-level indicators of perennial or intermittent reaches, but these families were not identified as family- or higher-level indicators of perennial or intermittent reaches.

The importance of taxa as predictors of streamflow-duration class varied between models balanced for flow class, but the most important indicators tended to vary less than those with higher minimum node depths in RF models (see Appendix S1 tabs 1–3). Differences in minimum node depth among taxa reflect the imbalance of streamflow classes across forests, zoogeographic distribution of invertebrate fauna, or differential responses to streamflow duration across ecoregions.

Table 4. Overall classification accuracy^a and classification accuracy by flow class (perennial, intermittent) and level II ecoregion (Eco) for the 9 streamflow-duration class balanced random forest models representing different taxonomic and numeric resolutions.

Taxonomic resolution	Numeric resolution	Classification accuracy (%)						
		Overall ($n = 370$)	Perennial ($n = 238$)	Intermittent ($n = 132$)	Eco 6.2 ($n = 33$)	Eco 6.2 ($n = 21$)	Eco 8.3 ($n = 81$)	Eco 8.4 ($n = 235$)
Aquatic family	Presence/absence	88.9	91.2	84.9	60.6	85.7	92.6	91.9
	Density	91.1	96.2	81.8	81.8	90.5	91.4	92.3
	Biomass	91.4	95.8	83.3	75.6	90.5	92.6	93.2
Aquatic + terrestrial family	Presence/absence	90.3	93.3	84.9	72.7	81.0	92.6	92.8
	Density	91.9	95.8	84.9	75.8	95.2	92.6	93.6
	Biomass	91.1	95.8	82.6	81.8	100.0	90.1	91.9
Aquatic genus	Presence/absence	92.4	95.4	87.1	81.8	100.0	95.1	92.3
	Density	93.2	96.6	87.1	84.9	100.0	96.3	92.8
	Biomass	91.6	95.8	84.1	78.8	90.5	93.8	92.8

^a % of reaches classified correctly by random forest analyses majority vote across 500 decision trees.

Table 5. Invertebrate taxa identified as indicators for perennial (P) and intermittent (I) reaches from random forest (RF) and indicator species analysis (ISA) for family- and higher-level (F) and genus-level (G) taxonomy datasets. Aquatic (A) and terrestrial (T) invertebrate families are labelled. – denotes that a taxon was not identified as an indicator of perennial or intermittent reaches. Taxa are ordered by highest to lowest median minimum node rank.

Taxonomic level	Taxon	Order	RF	ISA
F	Crangonyctidae A	Amphipoda	I	I
F	Elmidae A	Coleoptera	P	P
F	Ephemeridae A	Ephemeroptera	P	P
F	Enchytraeidae A	Enchytraeida	I	–
F	Isotomidae T	Collembola	I	I
F	Polycentropodidae A	Trichoptera	P	–
F	Dytiscidae A	Coleoptera	I	I
F	Lepidostomatidae A	Trichoptera	I	–
F	Ameletidae A	Ephemeroptera	–	I
F	Cecidomyiidae T	Diptera	I	I
F	Baetidae A	Ephemeroptera	P	P
F	Ephemerellidae A	Ephemeroptera	P	P
F	Psephenidae A	Coleoptera	P	P
F	Cicadellidae T	Hemiptera	I	I
F	Hydropsychidae A	Trichoptera	P	P
F	Capniidae A	Plecoptera	P	P
F	Cambaridae A	Decapoda	P	–
F	Corydalidae A	Megaloptera	P	P
F	Thremmatidae A	Trichoptera	I	I
F	Gomphidae A	Odonata	P	P
F, G	Megadrili A	Crassicitellata	I	I
F	Hydriphantidae A	Trombidiformes	–	I
F, G	Harpacticoida A	Copepoda	I	I
G	<i>Crangonyx</i>	Amphipoda	I	I
G	<i>Lirceus</i>	Isopoda	I	I
G	<i>Stempellinella</i>	Diptera	P	P
G	<i>Fridericia</i>	Enchytraeida	I	I
G	<i>Rheocricotopus</i>	Diptera	I	I
G	<i>Ephemera</i>	Ephemeroptera	P	P
G	<i>Chaetocladius</i>	Diptera	I	I
G	<i>Smittia</i>	Diptera	I	–
G	<i>Ameletus</i>	Ephemeroptera	–	I
G	<i>Polycentropus</i>	Trichoptera	P	P
G	<i>Heterosternuta</i>	Coleoptera	I	I
G	<i>Dicranota</i>	Diptera	–	P
G	<i>Nilotanytus</i>	Diptera	P	–
G	<i>Pseudorthocladius</i>	Diptera	I	–
G	<i>Isoperla</i>	Plecoptera	I	I
G	<i>Diplocladius</i>	Diptera	I	I
G	<i>Lepidostoma</i>	Trichoptera	I	–
G	<i>Ectopria</i>	Coleoptera	P	P
G	<i>Pristinella</i>	Tubificida	I	I
G	<i>Stenelmis</i>	Coleoptera	P	P
G	<i>Larsia</i>	Diptera	I	I
G	<i>Krenopelopia</i>	Diptera	I	–

Table 5. (Continued)

Taxonomic level	Taxon	Order	RF	ISA
G	<i>Limnophila</i>	Diptera	P	P
G	<i>Paraphaenocladus</i>	Diptera	I	–
G	<i>Brachypogon</i>	Diptera	I	–
G	<i>Mesocricotopus</i>	Diptera	I	I
G	<i>Oulimnius</i>	Coleoptera	P	–
G	<i>Paracladopelma</i>	Diptera	P	P
G	<i>Cryptochironomus</i>	Diptera	P	–
G	<i>Dipheter</i>	Ephemeroptera	P	–
G	<i>Heterotrissocladius</i>	Diptera	–	P
G	<i>Agabus</i>	Coleoptera	I	–
G	<i>Cryptolabis</i>	Diptera	P	P
G	<i>Parakiefferiella</i>	Diptera	–	P
G	<i>Ablabesmyia</i>	Diptera	P	–
G	<i>Eurylophella</i>	Ephemeroptera	–	P
G	<i>Stenacron</i>	Ephemeroptera	P	–
G	<i>Paracapnia</i>	Plecoptera	P	P
G	<i>Neophylax</i>	Trichoptera	I	I
G	<i>Nigronia</i>	Megaloptera	P	P
G	<i>Meropelopia</i>	Diptera	I	–
G	<i>Rheotanytarsus</i>	Diptera	–	P
G	<i>Baetis</i>	Ephemeroptera	P	P
G	<i>Diplectrona</i>	Trichoptera	P	P

Relative importance of season and habitat on RF predictions

Season was a more important factor than habitat in all RF models. The median rank node depth for habitat fell outside the invertebrate indicator threshold range ($\pm 20\%$ presence), whereas season fell within the range of the top invertebrate predictors for both the aquatic and terrestrial family-level datasets and the aquatic genus-level dataset (Fig. 2A, B).

ISA predictions of streamflow-duration class and comparisons with RF results

The taxa that ISA identified as predictive of streamflow-duration class were largely similar to those identified by RF models. ISA applied to family- or higher-level data identified 7 aquatic families and 3 terrestrial families as intermittent flow indicators and 9 aquatic families as perennial flow indicators (Table 5, Appendix S1 tab 2). For intermittent reaches, crangonyctid amphipods had high specificity (98%) and sensitivity (65%) relative to most indicator families. Among terrestrial indicator families, the cecidomyiids had the highest IV and sensitivity for intermittent reaches (68%). For perennial family indicators, elmids beetles had the highest IV and sensitivity (77%), whereas Ephemeroidea

had the highest specificity (100%). All families identified by ISA as perennial flow indicators were also identified as important perennial indicators by the RF models. However, RF models identified Cambaridae and Polycentropodidae as perennial flow indicators, but ISA did not. For intermittent flow indicators, ISA identified Ameletidae and Hydryphantidae, and RF models did not, whereas RF models identified Enchytraeidae and Lepidostomatidae, and ISA did not. Another 40 family- or higher-level taxa were identified as indicators by ISA when based on permutation testing (all p -value < 0.05) without restricting IV > 0.5 , but these did not include the 4 families identified only by RF (Appendix S1 tab 2).

For genus-level data, ISA and RF models jointly identified 13 aquatic genera as perennial flow indicators and 12 aquatic genera and 2 higher-level taxa (Harpacticoida and Megadrili) as intermittent flow indicators. ISA identified 18 aquatic genera as perennial flow indicators and 15 aquatic taxa (13 genera and 2 higher taxa) as intermittent flow indicators (Table 5, Appendix S1 tab 3). Genera from orders Ephemeroptera, Plecoptera, and Trichoptera were included among indicators of both intermittent and perennial reaches. Eleven indicator genera (6 perennial and 5 intermittent) were members of the ubiquitous family Chironomidae. *Stempellinella* spp. and *Rheocricotopus* spp. had the highest IV and sensitivity among perennial and intermittent genus

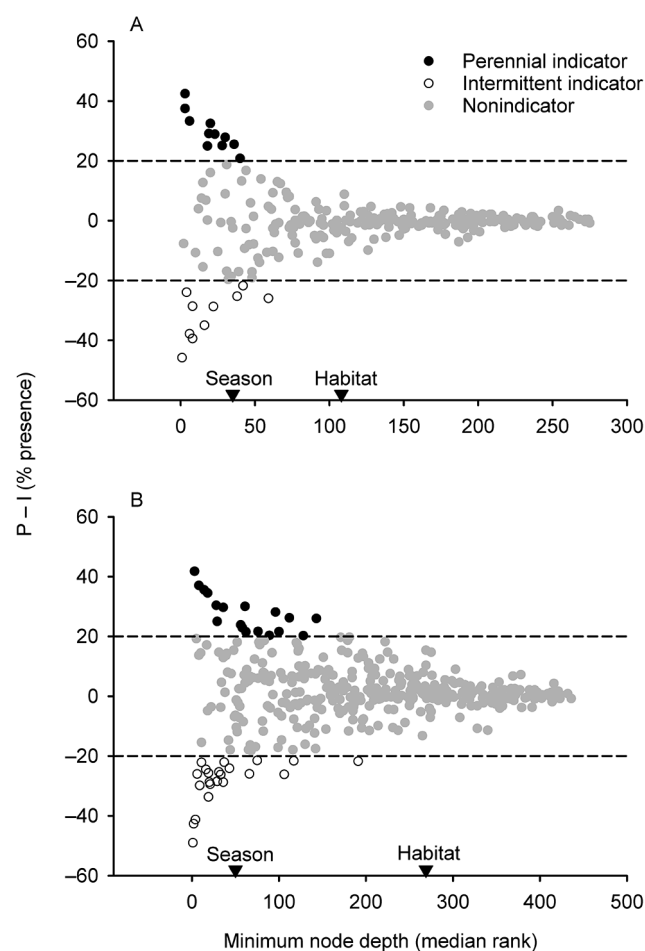


Figure 2. Identification of indicator taxa of perennial (P) and intermittent (I) streams based on random forest models and indicator species analysis at family-level and higher taxonomic resolution for aquatic and terrestrial life stages ($n = 217$ taxa) (A) and genus-level for aquatic taxa only ($n = 352$ taxa) (B). Each plot shows, for each taxon, the difference in % presence in samples from perennial and intermittent reaches relative to the median minimal node depth rank across the 3 random forest models (presence/absence, density, biomass). Horizontal dashed lines are the $\pm 20\%$ threshold criterion for identifying indicator taxa. The median rank of minimum node depth for season (wet and dry) and habitat type (erosional and depositional) is also shown.

indicators, respectively. All perennial flow indicators were aquatic insects having ≥ 1 terrestrial life stage, whereas 6 intermittent flow indicators were noninsects. Without limiting $IV > 0.5$, ISA identified another 104 indicators (all p -value < 0.05) but did not include 9/14 genera identified only by RF (Appendix S1 tab 3).

Variability in indicator taxa across ecoregions

Taxa identified as indicators of streamflow-duration class varied across level II ecoregions. Fifteen of 23 family-

or higher-level indicators and 16/44 genus-level indicators (from RF and ISA approaches combined) were collected from all study level II ecoregions (Fig. 3A, B). All indicator taxa identified were collected in ecoregion 8.4, whereas only 7 of the indicator family- or higher-level taxa and 19 of the genus-level indicators were collected in ecoregion 6.2. The mayfly genus *Ephemera* (family Ephemeridae), a perennial indicator, occurred in 37.8% of samples collected from perennial reaches and 0.8% of samples from intermittent reaches (Appendix S1 tab 3); however, this genus was only collected from forests in ecoregion 8.4 (Fig. 3A, B).

The probability that a reach was perennial tended to increase when a perennial indicator taxon was present and decreased when an intermittent indicator taxon was present, but these patterns varied across the study level II ecoregions (Fig. 3A, B). The probability of a reach being perennial in ecoregion 8.3 decreased in the presence of most perennial genus-level indicators but increased for the other study ecoregions. Similarly, the probability of a reach being perennial increased in ecoregion 6.2 in the presence of most intermittent indicators but decreased to varying extents for the other ecoregions. The dragonfly family Gomphidae was a perennial indicator across the entire dataset (occurring in 30.7% of perennial and 9.9% of intermittent reach samples); however, in ecoregion 5.3 this family was much more likely to occur in intermittent than perennial reaches (Fig. 3A). The isopod *Lirceus* was collected almost exclusively from intermittent reaches (occurring 44% of intermittent and only 1% of perennial samples) but was primarily collected in ecoregion 8.3 (96.7% of its occurrence). *Lirceus* was rarely collected in ecoregion 8.4 (3.3% of its occurrence) but only collected from perennial reaches, so its presence there had a positive change in probability of a reach being perennial (Fig. 3B). The differences in mean density and mean biomass of indicator taxa in perennial and intermittent reaches also varied among level II ecoregions (Figs 4A, B, 5A, B). Indicator taxa with highest overall relative occurrence in perennial or intermittent samples and IV values (Appendix S1 tabs 2, 3) tended to have lower among-ecoregion variation in differences in density and biomass between perennial and intermittent samples.

DISCUSSION

In this study we sought to identify indicator taxa that could be included in SDAMs for small, forested streams to determine whether numeric and taxonomic resolution affects classification accuracy for perennial and intermittent streamflow-duration classes and to assess the relative influence of season and habitat on classification accuracy. Although our primary objective was to identify indicators of intermittent and perennial reaches, we expected that these diverse, high-functioning stream ecosystems would support many of the same aquatic biota. Through our analysis of aquatic and terrestrial invertebrate samples from perennial

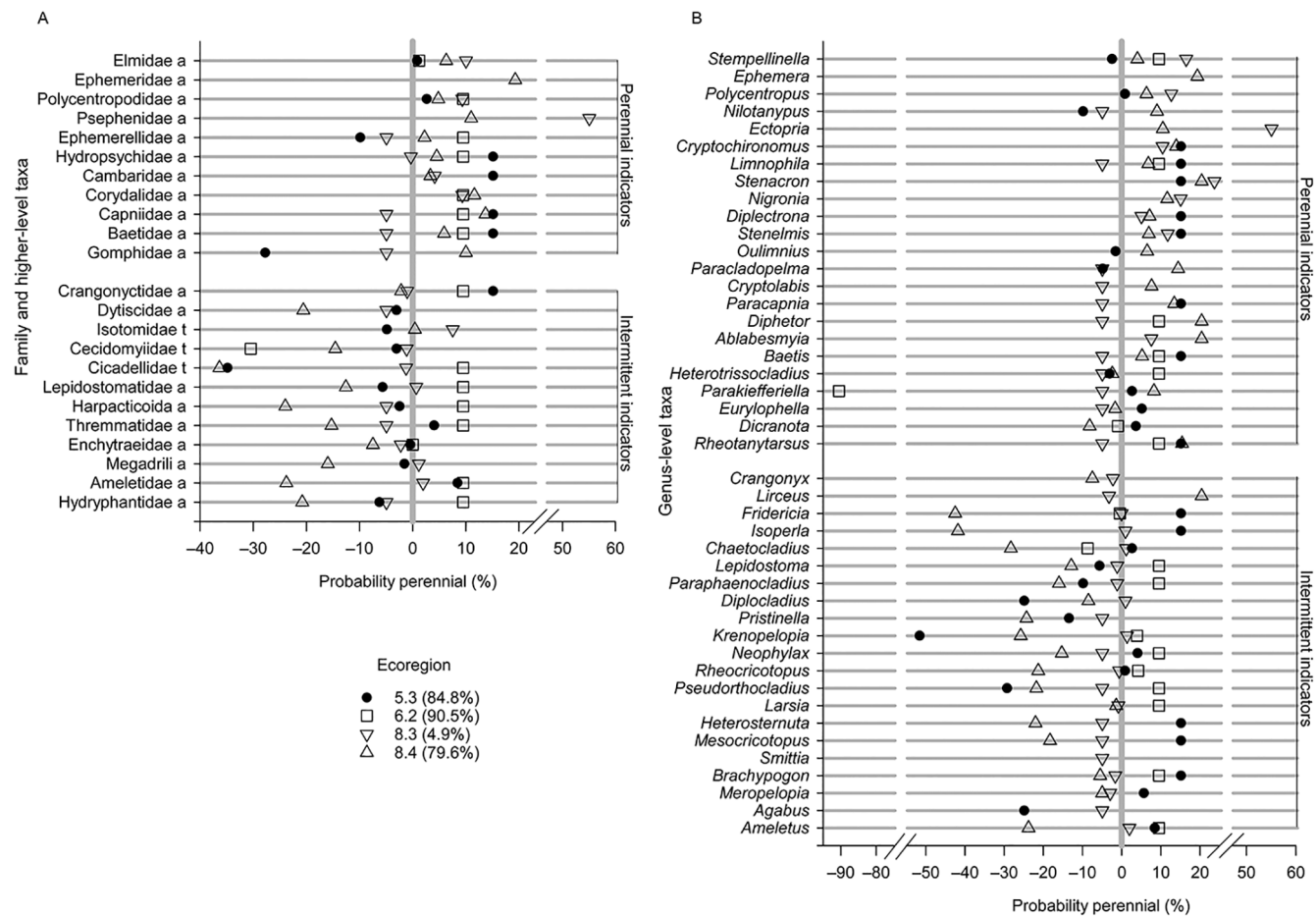


Figure 3. Change in probabilities of reaches being perennial based on presence of aquatic (a) and terrestrial (t) family-level and higher (A) and genus-level (B) perennial and intermittent indicators identified from random forest analyses and indicator species analyses at intermittent and perennial reaches by level II ecoregion. Ecoregion symbols show where taxa were collected. Symbols to the right of the vertical gray line indicate that reaches are more likely to be perennial. Symbols to the left of the vertical gray line indicate that reaches are less likely to be perennial (i.e., more likely to be intermittent). Percentage of samples collected from perennial reaches is shown in parentheses in the legend, which provides the baseline probability that a sample from an ecoregion is perennial (regardless of presence of a taxon) in this dataset. Note the breaks in panel A's and B's x-axes.

and intermittent reaches in headwater streams, we found, as expected, that most of the invertebrate taxa collected 1) were similarly represented between intermittent and perennial reaches or 2) occurred too rarely for use in indicator analysis (Fritz et al. 2020). Even so, the observed patterns of occurrence were sufficiently consistent within and distinct between intermittent and perennial reaches to identify groups of indicator taxa for each of the 2 streamflow-duration classes. Further, we found that the degree of consistency between the 2 approaches (RF and ISA) suggests that the identified indicator taxa may be useful in SDAMs, though there was geographic variation in the strength of association with streamflow-duration class for some indicators. Population measures at finer numerical resolution (biomass or density vs presence/absence) did not substantially improve classification accuracy, but identifying taxa to genus level rather

than family or higher level may be worth the additional effort, depending on geographic location and invertebrate family. In addition, including analysis based on invertebrate traits rather than taxonomy alone, as well as sampling across the seasons, may be useful.

Aquatic invertebrates are commonly used for monitoring the environmental status of rivers and streams, and numerous studies have developed invertebrate-based tools for measuring the effects of drought (Chadd et al. 2017), assessing reach flow status (Cid et al. 2016, England et al. 2019, Theodoropoulos et al. 2021), and detecting anthropogenic impacts across varying streamflow duration (Mazor et al. 2014, Soria et al. 2020). Because of their utility in discriminating among streamflow-duration classes, aquatic invertebrates are a component in most field-based SDAMs used in the United States (e.g., NCDWQ 2010,

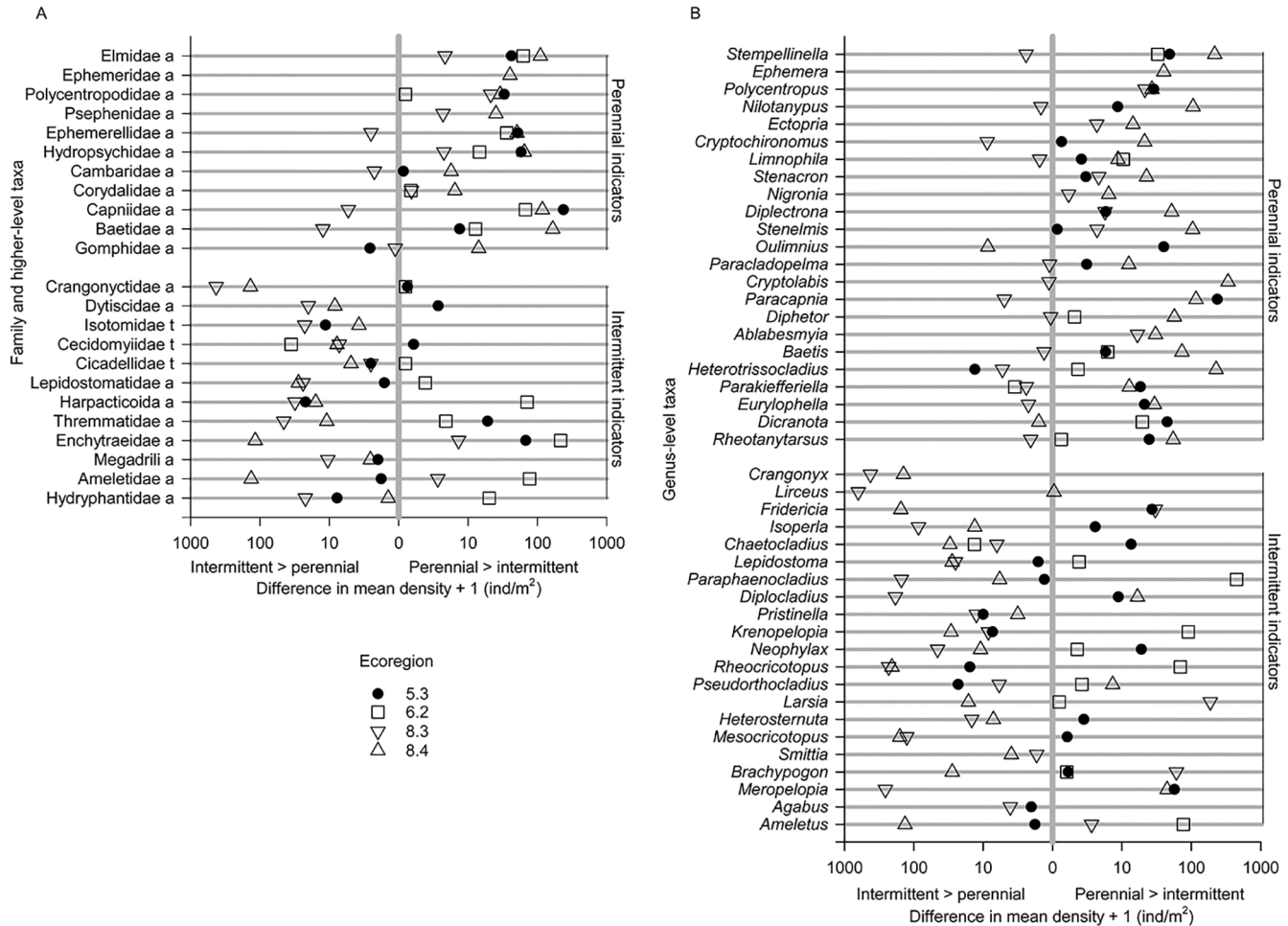


Figure 4. Difference in mean density (ind./m²) of aquatic (a) and terrestrial (t) family-level and higher (A) and genus-level (B) perennial and intermittent indicators identified from random forest analyses and indicator species analyses at perennial and intermittent reaches by level II ecoregion. Ecoregion symbols show where taxa were collected. Symbols to the right of the vertical gray line indicate mean density in samples from perennial reaches were higher than mean density in samples from intermittent reaches. Symbols to the left of the vertical gray line indicate mean density in samples from intermittent reaches were higher than mean density in samples from perennial reaches.

Nadeau 2015, OEPA 2020), the United Kingdom (Sarremejane et al. 2019), Mediterranean Europe (Prat et al. 2014), and Central Europe (Straka et al. 2021, Miliša et al. 2022). Most of the invertebrate taxa collected in the present study either occurred rarely or were similarly represented between intermittent and perennial reaches. When including ephemeral (surface flow only in direct response to precipitation), intermittent, and perennial reaches, aquatic macroinvertebrates have been among the most important indicators for field-based SDAMs (Fritz et al. 2013, Nadeau et al. 2015, Mazor et al. 2021a, b). This study excluded ephemeral reaches because the quantitative sampling method we used requires surface water, and the ephemeral reaches in the streams we studied often did not have surface water. Regardless, our findings suggest that there is rationale for incorporating invertebrate indicators of intermittent streamflow

into existing SDAMs, and we recommend further SDAM refinement and development.

Consistency in indicator taxa between methods and across ecoregions

Consistency of methods The 2 analytical approaches (RF, ISA) identified largely consistent lists of perennial and intermittent indicator taxa for forested headwater streams. There were 17 aquatic and terrestrial family- or higher-level taxa consistently identified by the RF and ISA approaches, including all 3 terrestrial family indicators of intermittent reaches. Of the 6 families identified by only 1 approach, Enchytraeidae, Polycentropodidae, Lepidostomatidae, and Cambaridae were identified by RF, and Ameletidae and Hydryphantidae were identified by ISA. There were 24

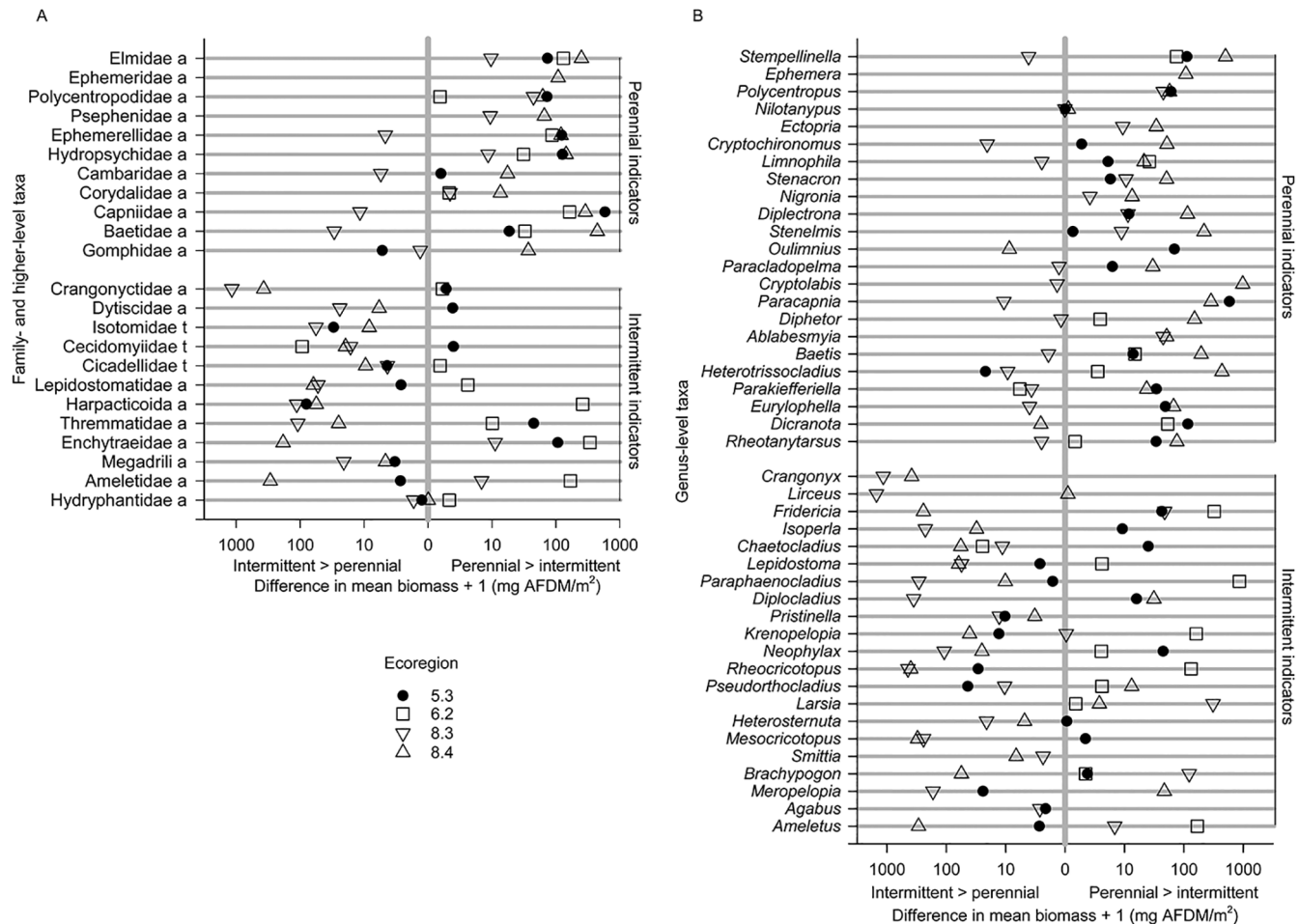


Figure 5. Difference in mean biomass as ash-free dry mass (mg AFDM/m²) of aquatic (a) and terrestrial (t) family-level and higher (A) and genus-level (B) perennial and intermittent indicators identified from random forest analyses and indicator species analyses at intermittent and perennial reaches by level II ecoregion. Ecoregion symbols show where taxa were collected. Symbols to the right of the vertical gray line indicate mean biomass in samples from perennial reaches were higher than mean biomass in samples from intermittent reaches. Symbols to the left of the vertical gray line indicate mean biomass in samples from intermittent reaches were higher than mean biomass in samples from perennial reaches.

aquatic genera consistently identified by the RF and ISA approaches: 12 indicators for intermittent reaches and 12 indicators for perennial reaches. *Crangonyx*, *Lirceus*, *Stempellinella*, and *Fridericia* had the highest median minimum node ranks and were identified as genus-level indicators by RF and ISA. Thirteen genera with the highest IV values were also identified as genus-level indicators using RF. Of the 20 genera identified by only 1 approach, 14 were by RF, and 6 were by ISA. *Ameletus* (the only genus in Ameletidae) was identified only by ISA as an indicator for intermittent reaches, whereas *Eurylophella*, *Dicranota*, and 3 chironomids (*Heterotrissocladius*, *Parakiefferiella*, and *Rheotanytarsus*) were identified only by ISA as indicators for perennial reaches. The 14 genera identified only by RF as indicators included 8 indicators for intermittent reaches and 6 indicators for perennial reaches. Taxa identified by both methods

may be considered more reliable than those identified by 1 method across our study sites. However, all taxa identified by 1 or both methods could be used in combination to explore indicator metrics (e.g., total number of indicator families for perennial reaches, total number of individuals from indicator genera for intermittent reaches).

Consistency across ecoregions As expected, the strength of association with streamflow-duration class for some indicators varied across level II ecoregions. Species within a genus or genera within a family may not have the same environmental preferences between different zoogeographic areas, resulting in contrasting distribution patterns for a genus or family between intermittent and perennial reaches (see Appendix S2, Tables S1, S2). Some variation in the strength of association among ecoregions can be attributed

to geographic differences in the occurrence (and, therefore, statistical representation) of flow classes. Additional variation is likely attributable to differences in adaptive responses to regional climate and landscape drivers of streamflow within families and genera (e.g., Lytle and Poff 2004). This geographic variability is illustrated by Miliša et al. (2022), who showed that the specific taxa associated with flow-duration classes also varied across continental, Mediterranean, and oceanic regions in Europe. In that study, all Elmidae (*Elmis*, *Esolus*, *Limnius*, and *Riolus* in continental; Elmidae in Mediterranean; and *Elmis* and *Riolus* in Oceanic) identified as indicator taxa from the 3 regions were restricted to groups of sites having mean annual flow duration of >90% of the year. In contrast, the amphipod genus *Niphargus* was the sole indicator for the lowest flow-intermittence group of the continental region, whereas another amphipod genus, *Gammarus*, was among the indicator taxa for >90%-flow groups for continental and oceanic regions (Miliša et al. 2022).

In the present study, the amphipod family Crangonyctidae was identified as the strongest intermittent indicator taxon but varied in its applicability across ecoregions. The genus *Crangonyx* was a strong indicator and dominant member of Crangonyctidae for ecoregions 8.3 and 8.4 but was not collected in ecoregions 5.3 or 6.2. *Stygobromus* was the only crangonyctid genus collected in ecoregions 5.3 and 6.2 and was not more prevalent or abundant in intermittent than perennial reaches. These results mirror a similar finding by Miliša et al. (2022) in which Simuliidae was identified as an indicator for the lowest-flow intermittent sites in the continental region but was among the indicator taxa for sites having mean annual flow duration of >90% in the oceanic and Mediterranean regions.

As with our findings, existing SDAMs in the United States differ in the specific taxa identified as indicators of streamflow-duration classes. The family-level indicators of perennial streams identified in the present study had high agreement with the perennial indicator-taxa lists identified in SDAMs previously developed for North Carolina (10/11 families, includes ecoregions 8.3 and 8.4; NCDWQ 2010) and Ohio (11/11 families, includes ecoregion 8.4; OEPA 2020). However, the family-level indicators of perennial streams that we identified had lower agreement with the taxa list for the Pacific Northwest SDAM (5/11 families, includes ecoregion 6.2; Mazzacano and Blackburn 2012, Nadeau 2015), which does not recognize any mayflies as being perennial indicator taxa. Most invertebrate-based indicators in existing SDAMs are specific to perennial reaches. Our findings show that invertebrate taxa indicative of intermittent reaches are as discriminatory as those for perennial reaches and, if included in SDAMs, could improve their classification accuracy. Our findings, along with these examples from the United States and Europe, support using region-specific indicator taxa for SDAM

metrics rather than using indicator lists over broader spatial scales.

Variable effects on indicator performance and classification accuracy

In addition to being a strong discriminator between streamflow-duration classes, an ideal indicator should be easily identified and measured, preferably while in the field and at any time of the year (e.g., presence/absence of a large-bodied invertebrate family commonly found in both wet and dry sampling seasons). To inform the level and type of effort needed, we evaluated the influence of numerical resolution and taxonomic level on the strength of associations between invertebrate taxa and 2 flow-duration classes in our analyses.

Numerical and taxonomic resolutions Classification accuracy was similar between models based on invertebrate density (91.1% overall accuracy) and presence/absence (88.9% overall accuracy). Accuracy was also comparable for analyses based on aquatic genus- and family-level data (92.4 and 88.9%, respectively). Previous studies have reported lower discriminatory power when using coarse numerical resolution but not taxonomic resolution (Marchant et al. 1995, Melo 2005); however, responses to environmental gradients can vary with taxonomic level because of differences in environmental preferences and behaviors among taxa within higher taxonomic levels and range distribution increases with taxonomic level (Jones 2008, Heino 2014).

Models based on estimated biomass did not improve classification accuracy for most study regions relative to presence/absence or density data. Two possibilities likely limit the added utility of biomass for flow classification compared with density or presence/absence. Because biomass can equivalently represent many small individuals or a few large individuals of a taxon, a taxon's biomass in a sample may not reflect the duration of time over which aquatic invertebrates have grown at the reach. In addition, our method for deriving biomass from body size may have limited our ability to detect more fundamental phenological differences within taxa that may occur between populations at intermittent and perennial reaches. Although use of allometric relationships can be more accurate than direct measurement because of mass loss through preservation, there is inherent variability associated with organisms that have instar development and with the lack of equations that account for variation across the geographic ranges of most taxa (Benke et al. 1999). Other means for characterizing size distributions (e.g., head-capsule width measurements) may reveal differences in life-history phenology, such as number of cohorts and degree of synchrony, that may differ between flow classes (e.g., Grubbs et al. 2006, DeJong and Canton 2013, Carey et al. 2021), but these more laborious techniques are infeasible for most SDAMs.

Although our data were based on a research study using quantitative samples that were sorted, identified, enumerated, and measured in the laboratory rather than derived from a timed field protocol, our findings suggest that SDAM macroinvertebrate indicators will more accurately distinguish intermittent and perennial flow classes if genus-level abundance data are used. However, the benefit from the added effort was modest for most of our study regions, and combining family-level identification with presence/absence data makes classification easier to implement for use in SDAMs (quicker, lower resource cost, less training; Marshall et al. 2006).

A notable exception to the use of family-level indicators is the family Chironomidae. Chironomids are among the most abundant, diverse, and widely distributed aquatic invertebrate groups in streams (Armitage et al. 1995), so their utility as an indicator at the family level is limited. In the present study, with 116 genera and comprising 51.9% of all individuals identified, Chironomidae was a diverse and dominant group with the most genus-level indicators (8 perennial and 10 intermittent) identified by 1 or both analytical methods (RF, ISA). Four chironomid genera (*Micropsectra*, *Heleniella*, *Tvetenia*, and *Thienemanniella*) were identified as regional indicators of perennial flow in understudied, spring-fed reaches in the Western Allegheny Plateau in ecoregion 8.4 (Pond et al. 2021). Other chironomid genera identified as indicators of perennial or intermittent flow in parts of the western and southwestern United States included *Stempellinella* and *Rheotanytarsus* (perennial indicators) and *Rheocricotopus* (an intermittent indicator). These perennial and intermittent indicator genera were among the chironomids found by Herbst et al. (2019) to decrease and increase, respectively, following drought in the California Sierra Nevada streams in ecoregion 6.2. Of the 41 drought-adapted genera identified in a review of chironomids in the American southwest by Cañedo-Argüelles et al. (2016), *Chaetocladius*, *Diplocladius*, *Larsia*, *Paraphaenocladius*, and *Smittia* were also identified as intermittent indicators in the present study. Additional chironomid indicator genera that corresponded between the 2 studies include *Ablabesmyia*, *Stempellinella*, and *Parakiefferella* for perennial reaches and *Chaetocladius* for intermittent reaches. However, important differences in our study results suggest the need for more study of the Chironomidae in this region. Of 20 chironomid genera identified by Cañedo-Argüelles et al. (2016) as indicators of streamflow duration (16 perennial and 4 nonperennial) in southeastern Arizona, only 5 were also identified as indicators in the present study with some conflicting indicator values. For example, Cañedo-Argüelles et al. (2016) identified the chironomid *Larsia* as a perennial flow indicator in southeast Arizona, whereas we identified *Larsia* as an intermittent flow indicator. *Parakiefferella* and *Rheotanytarsus*, also characterized as drought-adapted genera by Cañedo-Argüelles et al. (2016), were identified as perennial indicators in the present study.

Environmental factors Counter to our expectations, season (dry vs wet) ranked as a more important sampling factor in our dataset than habitat (erosional vs depositional). Because drying typically occurs more frequently, for longer durations, and more quickly in riffles than pools (Boulton 2003), we expected the habitat associations of invertebrates to be an important predictor of streamflow-duration class. Seasonality represents relatively predictable changes in potential environmental drivers (e.g., flow, temperature, and food) of aquatic invertebrate evolution (e.g., Lytle and Poff 2004) and is strongly associated with changes in assemblage composition over time (Johnson et al. 2012, Tonkin et al. 2017). The effect of seasonality may be especially strong when comparing assemblages from perennial and intermittent streams. Flow cessation and complete drying represent a strong environmental filter for the dry phase of intermittent reaches, whereas dispersal processes appear to be more important in structuring assemblages during the flowing phase (Datry et al. 2016, Sarremejane et al. 2017). Assemblages from pools and riffles of forested headwater streams in California were more similar during low and high flows (when pools and riffles, respectively, dominated reach habitat) than at moderate flows when pools and riffles were more evenly represented (Herbst et al. 2018). Feminella (1996) noted that some taxa collected in intermittent riffles but not in perennial riffles normally inhabit pools but had colonized drying riffles of forested streams in Alabama. Similarly, a mixture of rheophilic and lentic fauna were collected in experimental channels when surface flow was minimally connected between pools and riffles (Aspin et al. 2019). In another study, García-Roger et al. (2011) observed that microhabitat (e.g., dominant substrate type) explained slightly more variation in assemblage composition than season (wet vs dry), habitat type (riffle vs pool), or streamflow-duration class, although season–space interactions were evident. As the proportion of habitat types shifts with drying, season may be a more important consideration than habitat for using invertebrate indicators in SDAMs. We recommend sampling invertebrate indicators of perennial and intermittent reaches during seasons when most aquatic life stages of perennial and intermittent indicators are present and can be confidently identified. Although our findings support the currently recommended approach for most existing SDAMs of searching all available habitats for invertebrates and applying invertebrate indicators uniformly across all seasons, we add a cautionary note about the potential for an increasing influence of seasonality with climate change.

Traits of indicator taxa Analysis based on invertebrate traits rather than taxonomy can facilitate comparison of independent results across studies from different geographic locations (Poff et al. 2006). Whether aquatic invertebrates can persist in intermittent reaches or are restricted to perennial reaches may be related to their life histories and traits

conferring resistance or resilience to drying (Strachan et al. 2015). Resistance traits include presence of desiccation-resistant forms (e.g., cocoons, diapausing stages), body armoring that limits water loss, plastron or aerial respiration, and being limnophilous. Resilience traits include fast non-seasonal development, multiple generations per year (i.e., multivoltinism), and strong adult flight. To explore possible traits within our own taxa list, we assigned trait states from Poff et al. (2006) to indicator genera identified in the present study. We found some patterns consistent with our expectations, including more perennial indicators with semivoltine life cycles than intermittent indicators and more intermittent indicators with adult life stages able to exit water compared with perennial indicators. However, the 2 stream classes had comparable numbers of indicator taxa with several other functional traits, including slow seasonal development, presence of desiccation forms, no or poor body armor, plastron/aerial respiration, and being limnophilous.

Some studies have observed that trait patterns vary with hydrologic setting (Bogan et al. 2015, Crabot et al. 2021, Pond et al. 2021), but a recent long-term study on the effects of changing flow regimes that generalized life-history trait states did not predict most species' responses to flow class changing from perennial to intermittent (Carey et al. 2021). Taxa that appear to lack resistance or resilience to drying may still occur in intermittent reaches because they can rapidly recolonize from either nearby perennial reaches or a saturated hyporheic zone (Strachan et al. 2015, DeVecchia et al. 2022). In our study, regional differences in proximity to perennial reaches and extent of saturated hyporheic habitat may have contributed to the variable association strength between taxa and flow-duration class across level II ecoregions.

Incorporation of indicator taxa of intermittent streams into SDAMs

Most SDAMs used in the United States only consider the presence of perennial indicator taxa to inform streamflow-duration classification of reaches (NCDWQ 2010, Nadeau 2015, OEPA 2020), but as shown by our analyses, which identified a comparable number of aquatic indicator taxa for both intermittent and perennial reaches, identifying indicators of intermittent streams can be valuable for predicting streamflow-duration classes. Most studies have reported that taxa inhabiting intermittent reaches are a subset of the perennial reach assemblage (e.g., Arscott et al. 2010, Gauthier et al. 2020), but taxa that may be typically rare in perennial reaches may be dominant in intermittent reaches. For example, as drought was experimentally intensified in artificial channels in the southern United Kingdom, biting midges, including *Brachypogon* (an intermittent indicator taxon identified in the present study), became more dominant (Aspin et al. 2019). In the Czech Republic, aquatic Annelida (Oligochaeta) was a consistently dominant group before, during, and after the dry phase in 10 intermittent streams

(Pařil et al. 2019). In that study, 4 indicator taxa identified in the present study, *Eisenilla tetraedra* (Lumbricidae), *Fridericia* (Enchytraeidae), *Chaetocladius* (Chironomidae), and *Smittia* (Chironomidae), were among ~20 taxa that were collected alive after 100 d of dry conditions (Pařil et al. 2019). The enchytraeid *Fridericia* and the chironomid *Smittia* were also identified as indicators of ephemeral stream reaches in South Korea (Bae and Park 2019).

Additionally, most studies on stream invertebrate assemblages ignore terrestrial invertebrates found in aquatic samples, perhaps considering them uninformative. However, the stream bed of headwater streams can have emergent habitat, be near the stream banks or overhanging vegetation, and have tolerable conditions to terrestrial or semiterrestrial taxa (Steward et al. 2022). Drying should provide more habitat and opportunity for terrestrial fauna to occur on the stream bed of intermittent reaches than perennial reaches. Even though most terrestrial taxa were rare in this study and including terrestrial families in our models did not greatly improve classification accuracy compared with presence/absence models that used only aquatic families (90.3 vs 88.9% respectively), both RF and ISA approaches identified the same 3 terrestrial families as indicators of intermittent reaches. The terrestrial indicator families were the most frequently collected terrestrial taxa. Cecidomyiidae (45.7% of all samples), Isotomidae (29.4%), and Cicadellidae (18.6%) in our study have been documented by others as terrestrial or semi-aquatic insects that occur in nonperennial stream channels (Steward et al. 2017, England et al. 2019). However, most cecidomyiid larvae (98%) and isotomid collembolans (99%) and all cicadellids collected had body lengths <5 mm, making them less feasible than larger taxa as indicators of field-based SDAMs because small individuals are difficult to collect and identify under field conditions. Although not considered aquatic in most taxonomic guides (e.g., Merritt et al. 2019), 1 published aquatic invertebrate key recognizes some cecidomyiids as aquatic (Dobson 2013). Churchel and Batzer (2006) reported more cecidomyiids over the first 75 d of recovery in stream reaches that had completely dried than in reaches that retained surface water during a drought. Further study of terrestrial taxa is needed to identify indicators that are not only useful in distinguishing intermittent from perennial reaches but that may also prove to be adept at distinguishing ephemeral from intermittent reaches when channels are dry.

Recommendations for SDAM development

Our findings support several recommendations regarding the use of invertebrates in the development and refinement of SDAMs. First, the use of genus-level taxonomic resolution and density numeric resolution more clearly distinguished perennial from intermittent streams than did family-level and presence/absence resolution data, but the gain in discriminatory power was modest and may not be justified given

the added cost and level of effort. We therefore recommend incorporating categorical levels of abundance (e.g., none, 1–5, 5–10, >10 ind.) for invertebrate metrics in existing SDAMs. Second, given that season (wet vs dry) did not have a substantially stronger effect on discriminatory power than habitat (erosional vs depositional), we do not recommend more complex sampling or data adjustments than what is standard among existing SDAMs—i.e., careful sampling of the available habitat (flowing, standing, and dry in the assessment reach). Third, we recommend including terrestrial invertebrate indicators in addition to aquatic invertebrates for reach assessments. Finally, given that many potential indicators were not collected in all study regions or were equally useful in distinguishing perennial and intermittent reaches, we recommend development of region-specific indicator taxa for SDAMs rather than using indicator lists derived over broader geographic areas or predicted life-history and species traits. Although no single indicator is likely to provide a long-term characterization of hydrologic regimes, invertebrate indicator taxa should be part of a weight-of-evidence approach that uses multiple biological and physical parameters developed regionally to classify streamflow duration at the reach scale.

ACKNOWLEDGEMENTS

Author contributions: KMF, DMW, and BRJ conceived the study and collected field data. KMF coordinated the manuscript and led the writing. ROK, GJP, BJW, and KMF conducted data analysis. WTT and PCW conducted taxonomic analyses. All authors contributed to writing and provided editorial input.

We thank the United States Environmental Protection Agency (USEPA) Office of Wetlands, Oceans, and Watersheds; USEPA Regions 1–5, 9, and 10; the New York State Department of Environmental Conservation; Vermont Department of Environmental Conservation; the West Virginia Department of Environmental Protection; the United States Forest Service; the University of Kentucky; and the Nature Conservancy for logistical and field support. Karen Blocksom, Michael Griffith, and Thomas Barnum provided comments on a previous draft of the manuscript. Data are available from the lead author upon request.

The views expressed in this manuscript are those of the authors and do not necessarily reflect the views or policies of the USEPA. This manuscript has EPA tracking number ORD-049294. This research was subjected to US Geological Survey review and approved for publication. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the United States Government. All research was funded by the USEPA.

LITERATURE CITED

Allen, D. C., D. A. Kopp, K. H. Costigan, T. Datry, B. Hugueny, D. S. Turner, G. S. Bodner, and T. J. Flood. 2019. Citizen scientists document long-term streamflow declines in intermittent rivers of the desert southwest, USA. *Freshwater Science* 38:244–256.

Armitage, P. D., P. S. Cranston, and L. C. V. Pinder. 1995. *The Chironomidae: Biology and ecology of non-biting midges*. Chapman and Hall, London, United Kingdom.

Arscott, D. B., S. Larned, M. R. Scarsbrook, and P. Lambert. 2010. Aquatic invertebrate community structure along an intermittence gradient: Selwyn River, New Zealand. *Journal of the North American Benthological Society* 29:530–545.

Aspin, T. W. H., K. Hart, K. Khamis, A. M. Milner, M. J. O'Callaghan, M. Trimmer, Z. Wang, G. M. D. Williams, G. Woodward, and M. E. Ledger. 2019. Drought intensification alters the composition, body size, and trophic structure of invertebrate assemblages in a stream mesocosm experiment. *Freshwater Biology* 64:750–760.

Bae, M.-J., and Y.-S. Park. 2019. Evaluation of precipitation impacts on benthic macroinvertebrate communities at three different stream types. *Ecological Indicators* 102:446–456.

Benke, A. C., A. D. Huryn, L. A. Smock, and J. B. Wallace. 1999. Length-mass relationships for freshwater macroinvertebrates in North America with particular reference to the southeastern United States. *Journal of the North American Benthological Society* 18:308–343.

Bogan, M. T., K. S. Boersma, and D. A. Lytle. 2015. Resistance and resilience of invertebrate communities to seasonal and supra-seasonal drought in arid-land headwater streams. *Freshwater Biology* 59:491–501.

Boulton, A. J. 2003. Parallels and contrasts in the effects of drought on stream macroinvertebrate assemblages. *Freshwater Biology* 48:1173–1185.

Busch, M. H., K. H. Costigan, K. M. Fritz, T. Datry, C. A. Krabbenhoft, J. C. Hammond, M. Zimmer, J. D. Olden, R. M. Burrows, W. K. Dodds, K. S. Boersma, M. Shanafield, S. K. Kampf, M. C. Mims, M. T. Bogan, A. S. Ward, M. Perez Rocha, S. Godsey, G. H. Allen, J. R. Blaszcak, C. N. Jones, and D. C. Allen. 2020. What's in a name? Patterns, trends, and suggestions for defining non-perennial rivers and streams. *Water* 12:1980.

Cañedo-Argüelles, M., M. T. Bogan, D. A. Lytle, and N. Prat. 2016. Are Chironomidae (Diptera) good indicators of water scarcity? Dryland streams as a case study. *Ecological Indicators* 71:155–162.

Carey, N., E. T. Chester, and B. J. Robson. 2021. Life-history traits are poor predictors of species response to flow regime change in headwater streams. *Global Change Biology* 27:3547–3564.

Cartwright, J. M., K. A. Dwire, Z. Freed, S. J. Hammer, B. McLaughlin, L. W. Misztal, E. R. Schenk, J. R. Spence, A. E. Springer, and L. E. Stevens. 2020. Oases of the future? Springs as potential hydrologic refugia in drying climates. *Frontiers in Ecology and the Environment* 18:245–253.

Caton, L. W. 1991. Improving subsampling methods for the EPA “Rapid Bioassessment” benthic protocols. *Bulletin of the North American Benthological Society* 8:317–319.

Chadd, R. P., J. A. England, D. Constable, M. J. Dunbar, C. A. Extence, D. J. Leeming, J. A. Murray-Bligh, and P. J. Wood. 2017. An index to track the ecological effects of drought development and recovery on riverine invertebrate communities. *Ecological Indicators* 82:344–356.

Chapin, T. P., A. S. Todd, and M. P. Zeigler. 2014. Robust, low-cost data loggers for stream temperature, flow intermittency, and relative conductivity monitoring. *Water Resources Research* 50:6542–6548.

- Chen, C., A. Liaw, and L. Breiman. 2004. Using random forest to learn imbalanced data. Technical report. Department of Statistics, University of California, Berkeley, California.
- Churchel, M. A., and D. P. Batzer. 2006. Recovery of aquatic macroinvertebrate communities from drought in Georgia Piedmont headwater streams. *American Midland Naturalist* 156:259–272.
- Cid, N., I. Verkaik, E. M. García-Roger, M. Rieradevall, N. Bonada, M. M. Sánchez-Montoya, R. Gómez, M. L. Suárez, M. R. Vidal-Abarca, D. Demartini, A. Buffagni, S. Erba, I. Karaouzas, N. Skoulidakis, and N. Prat. 2016. A biological tool to assess flow connectivity in reference temporary streams from the Mediterranean Basin. 540:178–190.
- Corti, R., and T. Datry. 2012. Invertebrates and sestonic matter in an advancing wetted front travelling down a dry river bed (Albarine, France). *Freshwater Science* 31:1187–1201.
- Costigan, K. H., K. L. Jaeger, C. W. Goss, K. M. Fritz, and P. C. Goebel. 2016. Understanding controls on flow permanence in intermittent rivers to aid ecological research: Integrating meteorology, geology and land cover. *Ecohydrology* 9:1141–1153.
- Courtwright, J., and C. L. May. 2013. Importance of terrestrial subsidies for native brook trout in Appalachian intermittent streams. *Freshwater Biology* 58:2423–2438.
- Crabot, J., C. P. Mondy, P. Usseglio-Polatera, K. M. Fritz, P. J. Wood, M. J. Greenwood, M. T. Bogan, E. I. Meyer, and T. Datry. 2021. A global perspective on the functional responses of stream communities to flow intermittence. *Ecography* 44:1511–1523.
- Cutler, D. R., T. C. Edwards Jr, K. H. Beard, A. Cutler, K. T. Hess, J. Gibson, and J. J. Lawler. 2007. Random forests for classification in ecology. *Ecology* 88:2783–2792.
- Datry, T., N. Bonada, and J. Heino. 2016. Towards understanding the organisation of metacommunities in highly dynamic ecological systems. *Oikos* 125:149–159.
- Davis, J. A., A. Kerezy, and S. Nicol. 2017. Springs: Conserving perennial water is critical in arid landscapes. *Biological Conservation* 211:30–35.
- De Cáceres, M., F. Jansen, and N. Dell. 2019. *Indicspecies*: Relationship between species and group of sites. (Available from: <https://cran.r-project.org/web/packages/indicspecies/index.html>)
- De Cáceres, M., and P. Legendre. 2009. Associations between species and groups of sites: Indices and statistical inference. *Ecology* 90:3566–3574.
- DeJong, G. D., and S. P. Canton. 2013. Presence of long-lived invertebrate taxa and hydrologic permanence. *Journal of Freshwater Ecology* 28:277–282.
- DelVecchia, A. G., M. Shanafield, M. A. Zimmer, M. H. Busch, C. A. Krabbenhoft, R. Stubbington, K. E. Kaiser, R. M. Burrows, J. Hosen, T. Datry, S. K. Kampf, S. C. Zipper, K. Fritz, K. Costigan, and D. C. Allen. 2022. Reconceptualizing the hyporheic zone for nonperennial rivers and streams. *Freshwater Science* 41:167–182.
- Dobson, M. 2013. Family-level keys to freshwater fly (Diptera) larvae: A brief review and a key to European families avoiding use of mouthpart characters. *Freshwater Reviews* 6:2–32.
- Edwards, C. A. 1967. Relationships between weights, volumes, and numbers of soil animals. Pages 585–594 in O. Graff and J. E. Satchell (editors). *Progress in soil biology*. North Holland Publishing, Amsterdam, The Netherlands.
- England, J., R. Chadd, M. J. Dunbar, R. Sarremejane, R. Stubbington, C. G. Westwood, and D. Leeming. 2019. An invertebrate-based index to characterize ecological response to flow intermittence in rivers. *Fundamentals of Applied Limnology* 193:93–117.
- Epler, J. H. 2001. Identification manual for the larval Chironomidae (Diptera) of North and South Carolina: A guide to the taxonomy of the midges of the southeastern United States, including Florida. North Carolina Department of Environmental and Natural Resources, St. John's River Water Management District, Raleigh, North Carolina.
- Feminella, J. W. 1996. Comparison of benthic macroinvertebrate assemblages in small streams along a gradient of flow permanence. *Journal of the North American Benthological Society* 15:651–669.
- Fritz, K. M., N. Cid, and B. Autrey. 2017. Governance, legislation, and protection of intermittent rivers and ephemeral streams. Pages 477–507 in T. Datry, N. Bonada, and A. Boulton (editors). *Intermittent rivers and ephemeral streams*. Academic Press, London, United Kingdom.
- Fritz, K. M., B. R. Johnson, and D. M. Walters. 2006. Field operations manual for assessing the hydrologic permanence and ecological condition of headwater streams. EPA 600/R-06/126. United States Environmental Protection Agency, Office of Research and Development, Washington, DC. (Available from: https://www.epa.gov/sites/production/files/2015-11/documents/manual_for_assessing_hydrologic_permanence_-_headwater_streams.pdf)
- Fritz, K. M., B. R. Johnson, and D. M. Walters. 2008. Physical indicators of hydrologic permanence in forested headwater streams. *Journal of the North American Benthological Society* 27:690–704.
- Fritz, K. M., T.-L. Nadeau, J. E. Kelso, W. S. Beck, R. D. Mazor, R. A. Harrington, and B. J. Topping. 2020. Classifying streamflow duration: The scientific basis and an operational framework for method development. *Water* 12:2545.
- Fritz, K. M., W. R. Wenerick, and M. S. Kostich. 2013. A validation study of a rapid field-based rating system for discriminating among flow permanence classes of headwater streams in South Carolina. *Environmental Management* 52:1286–1298.
- García-Roger, E. M., M. M. Sánchez-Montoya, R. Gómez, M. L. Suárez, M. R. Vidal-Abarca, J. Latron, M. Rieradevall, and N. Prat. 2011. Do seasonal changes in habitat features influence aquatic macroinvertebrate assemblages in perennial versus temporary Mediterranean streams? *Aquatic Sciences* 73:567–579.
- Gauthier, M., B. Launay, G. Le Goff, H. Pella, C. J. Douady, and T. Datry. 2020. Fragmentation promotes the role of dispersal in determining 10 intermittent headwater stream metacommunities. *Freshwater Biology* 65:2169–2185.
- Grubbs, S. A., C. M. Thomas, B. T. Hutchins, and J. M. Taylor. 2006. Life cycles of *Allocaonia recta* and *Leuctra* spp. (Plecoptera: Capniidae and Leuctridae) across a flow gradient in a central Kentucky karst headwater stream. *Southeastern Naturalist* 5:321–332.
- Hafen, K. C., K. W. Blasch, A. Rea, R. Sando, and P. E. Gessler. 2020. The Influence of climate variability on the accuracy of NHD perennial and nonperennial stream classifications. *Journal of the American Water Resources Association* 56:903–916.

- Hammond, J. C., M. Zimmer, M. Shanafield, K. Kaiser, S. E. Godsey, M. C. Mims, S. C. Zipper, R. M. Burrows, S. K. Kampf, W. Dodds, C. N. Jones, C. A. Krabbenhoft, K. S. Boersma, T. Datry, J. D. Olden, G. H. Allen, A. N. Price, K. Costigan, R. Hale, A. S. Ward, and D. C. Allen. 2021. Spatial patterns and drivers of non-perennial flow regimes in the contiguous United States. *Geophysical Research Letters* 48:e2020GL090794.
- Hedman, E. R., and W. R. Osterkamp. 1982. Streamflow characteristics related to channel geometry of streams in Western United States. United States Geological Survey Water-Supply Paper 2193. United States Government Printing Office, Washington, DC. (Available from: <https://pubs.er.usgs.gov/publication/wsp2193>)
- Heino, J. 2014. Taxonomic surrogacy, numerical resolution and responses of stream macroinvertebrate communities to ecological gradients: are the inferences transferable among regions? *Ecological Indicators* 36:186–194.
- Herbst, D. B., S. D. Cooper, R. B. Medhurst, S. W. Wiseman, and C. T. Hunsaker. 2018. A comparison of the taxonomic and trait structure of macroinvertebrate communities between the riffles and pools of montane headwater streams. *Hydrobiologia* 820:115–133.
- Herbst, D. B., S. D. Cooper, R. B. Medhurst, S. W. Wiseman, and C. T. Hunsaker. 2019. Drought ecohydrology alters the structure and function of benthic invertebrate communities in mountain streams. *Freshwater Biology* 64:886–902.
- Hooshyar, M., S. Kim, D. Wang, and S. C. Medeiros. 2015. Wet channel network extraction by integrating LiDAR intensity and elevation data. *Water Resources Research* 51:10,029–10,046.
- Johnson, R. C., M. M. Carreiro, H.-S. Jin, and J. D. Jack. 2012. Within-year temporal variation and life-cycle seasonality affect stream macroinvertebrate community structure and biotic metrics. *Ecological Indicators* 13:206–214.
- Jones, F. C. 2008. Taxonomic sufficiency: The influence of taxonomic resolution on freshwater bioassessments using benthic macroinvertebrates. *Environmental Reviews* 16:45–69.
- Klemm, D. J., P. A. Lewis, F. Fulk, and J. M. Lazorchak. 1990. Macroinvertebrate field and laboratory methods for evaluating the biological integrity of surface waters. EPA/600/4-90/030. United States Environmental Protection Agency, Cincinnati, Ohio.
- Krabbenhoft, C. A., G. H. Allen, P. Lin, S. E. Godsey, D. C. Allen, R. M. Burrows, A. G. DelVecchia, K. M. Fritz, M. Shanafield, A. J. Burgin, M. Zimmer, T. Datry, W. K. Dodds, C. N. Jones, M. C. Mims, C. Franklin, J. C. Hammond, S. C. Zipper, A. S. Ward, K. H. Costigan, H. E. Beck, and J. D. Olden. 2022. Assessing placement bias of the global river gauge network. *Nature Sustainability* 5:586–592.
- Leigh, C. A. J. Boulton, J. L. Courtwright, K. Fritz, C. L. May, R. H. Walker, and T. Datry. 2016. Ecological research and management of intermittent rivers: An historical review and future directions. *Freshwater Biology* 61:1181–1199.
- Lytle, D. A., and N. L. Poff. 2004. Adaptation to natural flow regimes. *Trends in Ecology and Evolution* 19:94–100.
- Mahoney, D. T., J. R. Christensen, H. E. Golden, C. R. Lane, G. R. Evenson, E. White, K. M. Fritz, E. D'Amico, C. Barton, T. Williamson, K. Sena, and C. Agouridis. 2023. Dynamics of streamflow permanence in a headwater network: Insights from catchment-scale model simulations. *Journal of Hydrology* 620:129422.
- Marchant, R., L. A. Barmuta, and B. C. Chessman. 1995. Influence of sample quantification and taxonomic resolution on the ordination of macroinvertebrate communities from running waters in Victoria, Australia. *Marine and Freshwater Research* 46:501–506.
- Marshall, J. C., A. L. Steward, and B. D. Harch. 2006. Taxonomic resolution and quantification of freshwater macroinvertebrate samples from an Australian dryland river: The benefits and costs of using species abundance data. *Hydrobiologia* 572:171–194.
- Mazor, R. D., E. D. Stein, P. R. Ode, and K. Schiff. 2014. Integrating intermittent streams into watershed assessments: Applicability of an index of biotic integrity. *Freshwater Science* 33:459–474.
- Mazor, R. D., B. J. Topping, T.-L. Nadeau, K. M. Fritz, J. E. Kelso, R. A. Harrington, W. S. Beck, K. S. McCune, A. O. Allen, R. Leidy, J. T. Robb, and G. C. L. David. 2021a. Implementing an operational framework to develop a streamflow duration assessment method: A case study from the Arid West United States. *Water* 13:3310.
- Mazor, R. D., B. J. Topping, T.-L. Nadeau, K. M. Fritz, J. E. Kelso, R. A. Harrington, W. S. Beck, K. S. McCune, A. O. Allen, R. Leidy, J. T. Robb, G. C. L. David, and L. Tanner. 2021b. User manual for a beta streamflow duration assessment method for the western mountains of the United States. EPA-840-B-21008. United States Environmental Protection Agency, Washington, DC. P. 116. (Available from: <https://www.epa.gov/system/files/documents/2021-12/beta-sdam-for-the-wm-user-manual.pdf>, Accessed July 12, 2022).
- Mazzacano, C., and M. Blackburn. 2012. Macroinvertebrate indicators of streamflow duration for the Pacific Northwest: Companion field guide. Xerces Society for Invertebrate Conservation, Portland, Oregon. (Available from: https://www.epa.gov/sites/production/files/2015-10/documents/macroinvertebrate_field_guide.pdf, Accessed July 12, 2022).
- Melo, A. S. 2005. Effects of taxonomic and numeric resolution on the ability to detect ecological patterns at a local scale using stream macroinvertebrates. *Archiv für Hydrobiologie* 164:309–323.
- Merritt, R. W., and K. W. Cummins (editors). 1996. An introduction to the aquatic insects of North America. 3rd edition. Kendall Hunt, Dubuque, Iowa.
- Merritt, R. W., K. W. Cummins, and M. B. Berg (editors). 2019. An introduction to the aquatic insects of North America. 5th edition. Kendall Hunt, Dubuque, Iowa.
- Messenger, M. L., B. Lehner, C. Cockburn, N. Lamouroux, H. Pella, T. Snelder, K. Tockner, T. Trautmann, C. Watt, and T. Datry. 2021. Global prevalence of non-perennial rivers and streams. *Nature* 594:391–397.
- Miliša, M., R. Stubbington, T. Datry, N. Cid, N. Bonada, M. Šumanović, and D. Milošević. 2022. Taxon-specific sensitivities to flow intermittence reveal macroinvertebrates as potential bioindicators of intermittent rivers and streams. *Science of the Total Environment* 804:150022.
- Moon, H. P. 1956. Observations on a small portion of a drying chalk stream. *Proceedings of the Zoological Society of London* 126:327–334.
- More, A. S., and D. P. Rana. 2017. Review of random forest classification techniques to resolve data imbalance. Pages 72–78

- in 1st International Conference on Intelligent Systems and Information Management, Aurangabad, India, Institute of Electrical and Electronics Engineers.
- Moulton, S. R., J. L. Carter, S. A. Grotheer, T. F. Cuffney, and T. M. Short. 2000. Methods of analysis by the U.S. Geological Survey National Water Quality Laboratory – Processing, taxonomy, and quality control of benthic macroinvertebrate samples. United States Geological Survey Open-File Report 00-212. United States Geological Survey, Denver, Colorado.
- Nadeau, T.-L. 2015. Streamflow duration assessment method for the Pacific Northwest. EPA 910-K-14-001. United States Environmental Protection Agency, Region 10, Seattle, Washington. (Available from: https://www.epa.gov/system/files/documents/2022-03/sdam-pnw_nov-2015-final.pdf, Accessed July 12, 2022).
- Nadeau, T.-L., S. G. Leibowitz, P. J. Wigington, J. L. Ebersole, K. M. Fritz, R. A. Coulombe, R. L. Comeleo, and K. A. Blocksom. 2015. Validation of rapid assessment methods to determine streamflow duration classes in the Pacific Northwest, USA. *Environmental Management* 56:34–53.
- NCDWQ (North Carolina Division of Water Quality). 2010. Methodology for identification of intermittent and perennial streams and their origins. Version 4.11. North Carolina Department of Environment and Natural Resources, Division of Water Quality, Raleigh, North Carolina. (Available from: https://files.nc.gov/ncdeq/Water%20Quality/Surface%20Water%20Protection/401/Policies_Guides_Manuals/StreamID_v4point11_Final_sept_01_2010.pdf, Accessed July 12, 2022).
- Nolte, U. 1990. Chironomid biomass determination from larval shape. *Freshwater Biology* 24:443–451.
- OEPA (Ohio Environmental Protection Agency). 2020. Field methods for evaluating primary headwater habitat streams in Ohio. Version 4.1. Ohio Environmental Protection Agency, Division of Surface Water, Columbus, Ohio. (Available from: https://epa.ohio.gov/static/Portals/35/wqs/headwaters/PHWHManual_2020_Ver_4_1_May_2020_Final.pdf?ver=Jx6Z3rn9feBAUir3HWp_FQ%3d%3d, Accessed July 12, 2022).
- Omernik, J. M. 1995. Ecoregions: A framework for managing ecosystems. *The George Wright Forum* 12:35–50.
- Paluszynska, A., P. Biecek, and Y. Jiang. 2020. *randomForestExplainer*: Explaining and visualizing random forests in terms of variable importance. (Available from: <https://cran.r-project.org/web/packages/randomForestExplainer/index.html>)
- Pařil, P., M. Polářek, B. Loskotová, M. Straka, J. Crabot, and T. Datry. 2019. An unexpected source of invertebrate community recovery in intermittent streams from a humid continental climate. *Freshwater Biology* 64:1971–1983.
- Poff, N. 1996. A hydrogeography of unregulated streams in the United States and an examination of scale-dependence in some hydrological descriptors. *Freshwater Biology* 36:71–79.
- Poff, N. L., J. D. Olden, N. K. Vieira, D. S. Finn, M. P. Simmons, and B. C. Kondratieff. 2006. Functional trait niches of North American lotic insects: Traits-based ecological applications in light of phylogenetic relationships. *Journal of the North American Benthological Society* 25:730–755.
- Pond, G. J., K. J. G. Krock, and L. F. Ettema. 2021. Macroinvertebrates at the source: Flow duration and seasonality drive biodiversity and trait composition in rheocrene springs of the Western Allegheny Plateau, USA. *Aquatic Ecology* 56:99–121.
- Prat, N., F. Gallart, D. Von Schiller, S. Polesello, E. M. García-Roger, J. Latron, M. Rieradevall, P. Llorens, G. G. Barberá, D. Brito, A. M. De Girolamo, D. Dieter, A. Lo Porto, A. Buffagni, S. Erba, N. P. Nikolaidis, E. P. Querner, M. G. Tournoud, O. Tzoraki, N. Skoulidakis, R. Gómez, M. M. Sánchez-Montoya, K. Tockner, and J. Froebrich. 2014. The MIRAGE toolbox: An integrated assessment tool for temporary streams. *River Research and Applications* 30:1318–1334.
- Rosado, J., M. Morais, and K. Tockner. 2015. Mass dispersal of terrestrial organisms during first flush events in a temporary stream. *River Research and Applications* 31:912–917.
- Rosenberg, D. M., and V. H. Resh (editors). 1993. *Freshwater bio-monitoring and benthic macroinvertebrates*. Chapman and Hall, New York, New York.
- Sample, B. E., R. J. Cooper, R. D. Greer, and R. C. Whitmore. 1993. Estimation of insect biomass by length and width. *American Midland Naturalist* 129:234–240.
- Sánchez-Montoya, M. M., D. von Schiller, A. Ruhí, G. S. Pechar, L. Proia, J. Miñano, M. R. Vidal-Abarca, M. L. Suárez, and K. Tockner. 2016. Responses of ground-dwelling arthropods to surface flow drying in channels and adjacent habitats along Mediterranean streams. *Ecohydrology* 9:1376–1387.
- Sarremejane, R., M. Cañedo-Argüelles, N. Prat, H. Mykrä, T. Muotka, and N. Bonada. 2017. Do metacommunities vary through time? Intermittent rivers as model systems. *Journal of Biogeography* 44:2752–2763.
- Sarremejane, R., R. Stubbington, M. J. Dunbar, C. G. Westwood, and J. England. 2019. Biological indices to characterize community response to drying in streams with contrasting flow permanence regimes. *Ecological Indicators* 107:105620.
- Shanfield, M., S. A. Bourke, M. A. Zimmer, and K. H. Costigan. 2021. An overview of the hydrology of non-perennial rivers and streams. *WIREs Water* 8:e1504.
- Soria, M., C. Gutiérrez-Cánovas, N. Bonada, R. Acosta, P. Rodríguez-Lozano, P. Fortuño, G. Burgazzi, D. Vinyoles, F. Gallart, J. Latron, P. Llorens, N. Prat, and N. Cid. 2020. Natural disturbances can produce misleading bioassessment results: Identifying metrics to detect anthropogenic impacts in intermittent rivers. *Journal of Applied Ecology* 57:283–295.
- Stehr, F. W. (editor). 1987. *Immature insects*. Volume 1. Kendall/Hunt Publishing Company, Dubuque, Iowa.
- Stehr, F. W. (editor). 1991. *Immature insects*. Volume 2. Kendall/Hunt Publishing Company, Dubuque, Iowa.
- Steward, A. L., S. D. Langhans, R. Corti, and T. Datry. 2017. The biota of intermittent rivers and ephemeral streams: Terrestrial and semiaquatic invertebrates. Pages 245–271 in T. Datry, N. Bonada, and A. Boulton (editors). *Intermittent rivers and ephemeral streams*. Academic Press, London, United Kingdom.
- Steward, A. L., T. Datry, and S. D. Langhans. 2022. The terrestrial and semi-aquatic invertebrates of intermittent rivers and ephemeral streams. *Biological Reviews* 97:1408–1425.
- Strachan, S. R., E. T. Chester, and B. J. Robson. 2015. *Freshwater invertebrate life history strategies for surviving desiccation*. Springer Science Reviews 3:57–75.
- Straka, M., M. Polářek, Z. Csabai, O. Zweidick, G. Wolfram, E. I. Meyer, E. M. Elexová, M. Leřtáková, and P. Pařil. 2021. Stream drying bioindication in Central Europe: A biodrought index accuracy assessment. *Ecological Indicators* 130:108045.

- Stubbington, R., M. T. Bogan, N. Bonada, A. J. Boulton, T. Datry, C. Leigh, and R. Vander Vorste. 2017. The biota of intermittent rivers and ephemeral streams: Aquatic invertebrates. Pages 217–243 *in* T. Datry, N. Bonada, and A. Boulton (editors). Intermittent rivers and ephemeral streams. Academic Press, London, United Kingdom.
- Theodoropoulos, C., I. Karaouzas, and R. Stubbington. 2021. Biotic indices of hydrological variability as tools to inform dynamic ecological status assessments in river ecosystems. *Journal of Environmental Management* 295:113124.
- Thorp, J. H., and A. P. Covich (editors). 2001. Ecology and classification of North American freshwater invertebrates. 2nd edition. Academic Press, San Diego, California.
- Tonkin, J. D., M. T. Bogan, N. Bonada, B. Rios-Touma, and D. A. Lytle. 2017. Seasonality and predictability shape temporal species diversity. *Ecology* 98:1201–1216.
- van Meerveld, H. J. I., E. Sauquet, F. Gallert, C. Sefton, J. Seibert, and K. Bishop. 2020. *Aqua temporaria incognita*. *Hydrological Processes* 34:5704–5711.
- Ward, A. S., N. M. Schmadel, and S. M. Wondzell. 2018. Simulation of dynamic expansion, contraction, and connectivity in a mountain stream network. *Advances in Water Resources* 114:64–82.
- Williams, D. D. 2006. *The biology of temporary waters*. Oxford University Press, Oxford, United Kingdom.
- Woelfle-Erskine, C. 2017. Collaborative approaches to flow restoration in intermittent salmon-bearing streams: Salmon Creek, CA, USA. *Water* 9:217.
- Wright, M. N., S. Wager, and P. Probst. 2022. *ranger*: A fast implementation of random forests. (Available from: <https://cran.r-project.org/web/packages/ranger/ranger.pdf>)